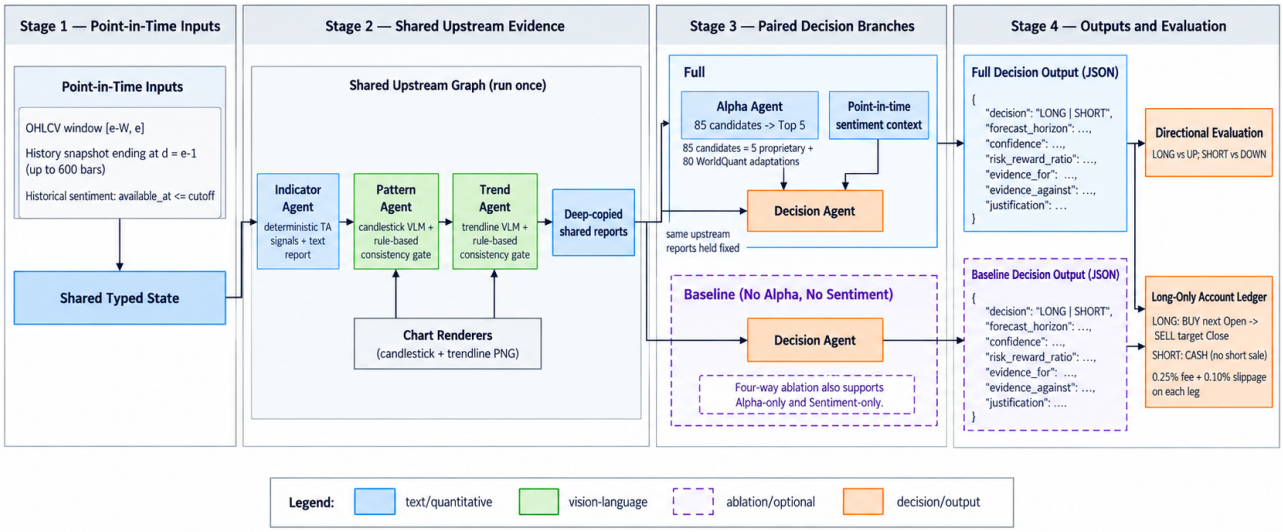
Graphical Abstract

Evaluating Quantitative Evidence in Multi-Agent Decision Support for Vietnamese Equities*

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Highlights

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- A class-imbalance audit qualifies the legacy multi-agent benchmark.
- Constant CASH exceeds legacy Full in raw accuracy and mean return.
- Train-fitted numerical baselines use explicit temporal preprocessing.
- Snapshot differences prevent a valid retrospective Full comparison.
- The available evidence does not establish alpha value or profitability.

Evaluating Quantitative Evidence in Multi-Agent Decision Support for Vietnamese Equities

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Abstract

We examine quantitative evidence in a five-agent decision-support architecture for Vietnamese equities using an audit of archived predictions and an explicitly separate numerical-baseline reconstruction. The legacy benchmark contains 180 decisions across nine equities. Its Full branch achieves 65.0% accuracy, below a constant DOWN/CASH rule at 68.9%; balanced accuracy is 55.0% and Matthews correlation is 0.113. The Base reports control, which removes both factors and sentiment, has balanced accuracy of 44.4%. Mean independently compounded Full account return is negative, whereas CASH returns zero. Consequently, the Full–Base accuracy gap does not establish alpha information value or profitability. Additional train-fitted majority, momentum, logistic and factor-only baselines, together with continuous-hold and daily mark-to-market ledgers, are evaluated on a newly retrieved, frozen price snapshot. That snapshot differs from legacy at 74 of the 180 cycles and changes two labels, so these numerical results are not presented as a head-to-head comparison with archived Full predictions. Missing historical reports and training predictions also prevent verification of an identical-information comparator and a Full-exposure-matched control. The reanalysis documents class imbalance, target definitions, preprocessing boundaries and the remaining evidence needed before claiming predictive superiority.

Keywords: Multi-agent LLM, Knowledge fusion, Knowledge integration, Financial decision support, Alpha factor selection

1. Introduction

Large language models (LLMs) have recently shown strong potential as reasoning engines in financial analysis. Their value for knowledge-based decision support depends on reliable integration of heterogeneous quantitative, textual, and visual evidence; distributing tasks among agents alone is insufficient. Frameworks such as QuantAgent [1] provide a compelling blueprint for LLM-driven trading, but they primarily address mature, English-language markets and leave the knowledge-integration requirements of emerging economies underexplored.

Vietnam's equity market presents three challenges that require explicit adaptation. First, a predictive horizon is defined over future bars, whereas an executable cash-equity simulation depends on explicit entry, exit, and position assumptions. Calendar gaps and instrument scope further prevent

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a bar horizon from being interpreted automatically as an equivalent economic holding period. Second, the dominant financial news source, CafeF, publishes exclusively in Vietnamese, making English-centric sentiment pipelines inapplicable. Third, the shallow cross-sectional universe of HOSE, HNX, and UPCoM calls for alpha factor selection methods suited to single-stock Open, High, Low, Close, and Volume (OHLCV) data instead of the large-universe assumptions of standard factor libraries.

We present a five-agent LangGraph [2] architecture that addresses these gaps through explicit knowledge fusion. An Indicator Agent supplies rule-based technical evidence; an Alpha Agent combines point-in-time selection from 85 OHLCV candidates with separately represented Vietnamese sentiment context; Pattern and Trend agents analyse complementary chart views; and a Decision Agent integrates the resulting reports into a binary LONG/SHORT forecast whose account action is resolved separately. This architecture provides a reproducible basis for integrating heterogeneous quantitative, textual, and visual evidence in an emerging-market decision-support setting. Section 4 documents sentiment construction and provenance controls at the methodological level.

The objective is to evaluate the evidence required to establish whether the additional quantitative and sentiment bundle contributes predictive information. The legacy protocol shares Indicator, Pattern and Trend reports within each Full–Base pair, but the control removes both factors and sentiment; their accuracy gap is therefore a bundle comparison, not isolated alpha attribution. Moreover, CASH exceeds Full in raw accuracy and mean return. We reanalyse class balance and both classes' precision/recall, and evaluate numerical baselines with temporal train/validation preprocessing on a separately frozen reconstruction. Price differences and missing historical inputs prevent a valid identical-information comparison with all archived Full decisions. The four-way legacy experiment is retained as an unverified diagnostic, not evidence closing this attribution gap.

The main contributions are as follows:

- We describe the five-agent implementation and distinguish its information bundle from the Base reports control without claiming isolated component value.
- An independent legacy audit reports class support, confusion matrices, balanced accuracy, MCC and both classes' precision/recall alongside constant baselines.
- A numerical baseline pipeline fits preprocessing, model parameters and decision thresholds on chronological train/validation data, while explicitly testing snapshot comparability.
- Separate cycle and daily ledgers distinguish CASH, repeated LONG cycles and continuous holding; the resulting evidence does not establish predictive superiority or deployable profitability.

The remainder of this paper is organised as follows. Section 2 reviews related work on LLM-based trading agents, Vietnamese NLP, and quantitative alpha research. Section 3 formalises the problem. Section 4 describes the system architecture. Section 5 presents the experimental setup. Section 6 reports backtest results. Section 7 discusses limitations, and Section 8 concludes.

2. Related works

2.1. LLM Agents in Financial Analysis

The application of LLMs to financial tasks has grown rapidly, spanning earnings call analysis, financial question answering, and report summarisation. FinGPT [3] and BloombergGPT [4] demonstrate that domain-adapted language models outperform general-purpose ones on financial benchmarks. More relevant to our work, TradingAgent [5] and QuantAgent frame trading as a multi-step

reasoning problem, delegating sub-tasks-market perception, reflection, and decision-to coordinated LLM agents. Our framework builds directly on this paradigm but extends it with vision-language chart analysis, quantitative alpha integration, and emerging-market adaptations absent from prior systems.

2.2. Technical Analysis and Pattern Recognition

Automating technical analysis has been explored through convolutional neural networks for candlestick pattern recognition [6], reinforcement learning for indicator-based policy learning [7], and hybrid rule-based systems [8]. Recent work has begun replacing handcrafted features with multimodal chart-understanding models. ChartLlama [9] demonstrates broad chart-description capability, but generic chart competence does not establish faithful use of financial-chart evidence. FinChart-Bench finds material instruction-following and spatial-reasoning limitations across 26 vision-language models on financial charts [10]. More specifically, the Martingale Doppelgänger-Eval study shows that observational chart accuracy cannot identify whether a model used local candlestick evidence or merely extrapolated the prevailing trend; matched evidence interventions are required to separate these behaviours [11]. Our two vision agents therefore use deterministic contradiction checks as a safety gate, but we do not interpret that gate as a causal evidence-faithfulness audit.

2.3. Quantitative Alpha Factor Research

Systematic factor construction has a long tradition in quantitative finance, from foundational momentum and value factors [12, 13] to the large-scale formulaic approach of Kakushadze, who codified 101 analytic alpha expressions (WorldQuant 101). Subsequent work adapts formulas to single-asset settings [14] and evaluates factor decay and capacity in different market regimes [15, 16]. Factor mining is nevertheless susceptible to data-snooping and multiplicity: selecting and assessing a model on the same resample can bias its apparent error, whereas an outer validation loop must repeat the complete inner selection procedure [17]. Financial strategy selection adds non-normal returns and the number of trials, which motivate the Deflated Sharpe Ratio [18]. Our executable registry contains five proprietary formulas and 80 adaptations selected from the 101-formula source catalogue. We dynamically select five candidates per cutoff, but the reported experiment does not use nested validation or a Deflated Sharpe correction; its candidate-level IC and Sharpe values must therefore be read as selection diagnostics rather than confirmatory evidence for individual factors.

2.4. Sentiment Analysis in Finance and Vietnamese NLP

News sentiment has been established as a statistically significant predictor of short-term price movements [19], with transformer-based models such as FinBERT [20] setting the standard for English financial text. For Vietnamese, ViSoBERT provides a BERT-based model pre-trained on large-scale Vietnamese corpora and achieves strong results on Vietnamese sentiment classification benchmarks. Financial-domain validity does not follow automatically from general-language benchmark performance. We therefore use the public `5CD-AI/Vietnamese-Sentiment-visobert` classifier through a local Transformers backend by default, with an optional Hugging Face Dedicated Inference Endpoint and an explicit lexicon fallback. The actual scoring path is recorded for every newly scored article. Target and related-company sentiment are presented in a separate LLM text context and do not enter the OHLCV factor formulas or ranking. Legacy cache entries that lack scorer metadata are treated as unknown rather than retrospectively attributed to ViSoBERT.

2.5. Trading Systems for Emerging Markets

Research on algorithmic trading in emerging markets is comparatively sparse. Studies on the Vietnamese market have primarily examined market efficiency [21], herding behaviour [22], and the predictive power of classical technical indicators under the price-limit regime [23], but none employs LLM-based agents. Our work develops a multi-agent knowledge-fusion and decision-support framework that records Vietnamese cash-equity execution and settlement assumptions separately from the domestic news ecosystem and single-stock OHLCV data characteristics. Consequently, the model horizon remains an explicit experimental choice instead of being inferred from a universal settlement rule.

2.6. Evaluation Standards and Agentic Benchmarks

Recent literature highlights the critical need for rigorous, standardized evaluation in LLM-driven trading. For instance, FinVision [24] shares our modular multi-agent approach and vision-based technical analysis but places much stronger emphasis on extensive ablations (e.g., of reflection modules) and comprehensive risk reporting (Sharpe, drawdowns). Furthermore, an audit-oriented survey on financial LLMs [25] identifies widespread methodological gaps in time-consistent splits, transaction-cost modeling, and reproducibility. While our framework addresses costs and lookahead leakage, we recognize that our reporting detail currently falls short of the rigorous bar for full reproducibility. Finally, live/prediction-market benchmarks [26, 27] and forward-looking evaluation paradigms like FutureX [28] underscore that static walk-forward backtests–while useful–must ultimately be complemented by live trials and execution realism to substantiate strong claims about agentic efficacy.

3. Problem Formulation

3.1. Market Model and Data

Let $\mathcal{S} = \{s_1, s_2, \ldots, s_N\}$ denote the universe of Vietnamese equities listed on HOSE, HNX, or UPCoM. For each asset $s \in \mathcal{S}$, the market is observed as a discrete-time OHLCV series:

$$\mathbf{x}_u = \big(O_u,\, H_u,\, L_u,\, C_u,\, V_u\big) \;\in\; \mathbb{R}_+^5, \quad u = 0, 1, \ldots, T-1, \tag{1}$$

where O_u, H_u, L_u, C_u, V_u denote the open, high, low, closing price, and trading volume at zero-based bar position u respectively. When volume observations are missing for more than 50% of bars, the intra-bar range $H_u - L_u$ is substituted as a liquidity proxy.

We use zero-based row positions and an exclusive window-end boundary. For a test point with boundary e, the analysis window of size W is exactly:

$$\mathbf{X}_e = \big(\mathbf{x}_{e-W},\, \mathbf{x}_{e-W+1},\, \ldots,\, \mathbf{x}_{e-1}\big) \;\in\; \mathbb{R}^{W\times 5}. \qquad (\text{equivalently, } \mathbf{X}_e = \mathbf{X}_{[e-W,e)}) \tag{2}$$

Thus e is not itself a visible bar: the final visible (decision) bar is $d = e - 1$. The next bar, e, is reserved for the earliest simulated entry, and the immutable label target is $h = e-1+L$. A candidate economic exit may use that close only if the separate execution contract accepts it. This mapping is used in the engine, alpha labels, persisted provenance, and the experiments:

$$\boxed{\text{window} = [e-W, e), \qquad d = e-1, \qquad \text{entry} = e, \qquad h = e-1+L.} \tag{3}$$

The mapping is positional rather than calendar arithmetic; timestamps are carried alongside every index, so a weekend or holiday gap does not create a synthetic bar. The first complete window is $e = W$, and it is evaluable only when the target is present, i.e., $T \geq W + L$. For $L = 1$, the target is the entry bar ($h = e$), while for $L = 3$ it is two positions after entry ($h = e + 2$). Any boundary whose target is outside the available frame is rejected rather than shortened.

Algorithm 1: Walk-Forward Evaluation

Input: OHLCV series $\mathbf{X}$, alpha registry $\mathcal{A}$, historical sentiment cache $\mathcal{D}$, window size W, lookahead L, step size S

Output: Directional predictions $\hat{y}_{e,L}$, accuracy, Alpha Lift, and compounded account return

Initialize results list $\mathcal{R} \leftarrow \emptyset$

foreach *test point e in the walk-forward schedule* **do**

- Extract the exact analysis window $\mathbf{X}_e = \mathbf{X}_{[e-W,e)}$
- Compute technical indicators using TA-Lib
- Evaluate all candidate alphas $\alpha^{(k)} \in \mathcal{A}$
- Remove candidates with no fitted transform, fewer than 20 finite evaluation observations, or non-finite ranking metrics to form $\mathcal{V}$
- Min–max normalise each metric over $\mathcal{V}$ and rank using:

$$S_k = 0.40\mathcal{N}_{\mathcal{V}}(|IC_k|) + 0.35\mathcal{N}_{\mathcal{V}}(Acc_k) + 0.25\mathcal{N}_{\mathcal{V}}(Sharpe_k)$$

- Select top-5 alpha factors $\mathcal{A}^*$
- Retrieve historical CafeF sentiment at the decision cutoff $d = e - 1$
- Generate candlestick and trendline chart images from the same exact $\mathbf{X}_e$ window
- Run the Indicator, Alpha, Pattern, and Trend Agents
- Pass all reports to Decision Agent
- Obtain binary prediction $\hat{y}_{e,L} \in \{\texttt{LONG}, \texttt{SHORT}\}$
- Compare prediction with realised return
- Store metrics in $\mathcal{R}$

Compute directional accuracy and Alpha Lift

Compute compounded account return under Eq. (34)

return $\mathcal{R}$

3.2. Directional Prediction Task

The core task predicts whether a candidate LONG cycle is profitable after the configured transaction costs. With $d = e - 1$ and $h = e - 1 + L$, define

$$r_{e,L}^{\text{net}} = \frac{C_h}{O_e}\frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1, \qquad y_{e,L} = \begin{cases} +1, & r_{e,L}^{\text{net}} > 0, \\ -1, & r_{e,L}^{\text{net}} \le 0. \end{cases} \tag{4}$$

Here O_e is the next-bar entry open and C_h is the target close. The same net Open-to-Close return defines the directional label, the alpha-selection target, and the realised return of an executed LONG. Thus an opening gap and both transaction legs are represented consistently. The reported study uses daily bars with $L = 3$; intraday execution is outside its empirical scope.

A trading system $\mathcal{F}$ maps each analysis window to a binary decision:

$$\hat{y}_{e,L} \;=\; \mathcal{F}\big(\mathbf{X}_e\big) \;\in\; \{-1, +1\}, \tag{5}$$

where $+1$ corresponds to a `LONG` signal and -1 to `SHORT`.

For the reanalysis, the net economic target above is retained unchanged. A separate secondary target is $y^{\text{gross}} = +1$ if $C_{e-1+L}/O_e - 1 > 0$, and -1 otherwise. Gross-target metrics, where reported in the reanalysis package, use the same predictions and are not used to retune models or replace the net target. Thus the reported task is profitable-cycle classification, rather than an unqualified forecast of the sign of the close-to-close price change.

3.3. Alpha Factor Formalisation and Adaptation

An *alpha factor* $\alpha^{(k)}$ is a measurable function that maps a feature frame derived from $\mathbf{X}_e$ to a scalar signal:

$$\alpha^{(k)} : \mathbb{R}^{T_{\text{hist}} \times d} \longrightarrow \mathbb{R}, \tag{6}$$

where d is the number of engineered features and T_{hist} is the historical window used for computation.

The executable registry contains 80 adaptations drawn from the WorldQuant-101 catalogue, not all 101 source expressions, plus five proprietary formulas. The source expressions were designed primarily for cross-sectional ranking across a broad, liquid equity universe. To apply the registered subset to a single Vietnamese stock, we use a *semantic translation layer* that replaces cross-sectional operators with rolling time-series surrogates. This is an engineering heuristic rather than an invariance-preserving mathematical transformation.

Operator Localisation.. The primary transformation involves the cross-sectional rank(x) operator, which denotes the relative percentile of a stock against its peers. A cross-sectional rank is localised to a 20-bar time-series percentile rank unless the source expression already specifies a time-series rank with its own window. Let $\mathcal{H}_{u,w}$ contain the available preceding observations in that rolling window and let $n_{u,w} = |\mathcal{H}_{u,w}| + 1$. The implementation computes

$$\text{ts_rank}(x_u, w) \;=\; \frac{1}{\max(n_{u,w} - 1, 1)} \sum_{x_v \in \mathcal{H}_{u,w}} \mathbf{1}[x_u > x_v]. \tag{7}$$

Thus ties are not counted as strictly lower values, and an initial partial window is permitted once the operator-specific minimum is met. Other explicit time-series lookbacks are retained from each registered expression and can be substantially longer than 20 bars. Average Daily Volume (ADV_n) is the asset's own n-day rolling mean volume, and VWAP is approximated by the daily typical price $(H_u + L_u + C_u)/3$. Functions $\delta(x, d)$ and delay(x, d) are implemented as d-period differences and lags. These substitutions remove peer-relative information, so neither the economic interpretation nor the stochastic properties of the source cross-sectional alpha are presumed to survive unchanged.

Consistent Normalisation.. To ensure comparability across candidates with different scales, each adapted raw alpha value is normalised to $[-1, 1]$ via:

$$\tilde{\alpha}_{u,e}^{(k)} \;=\; \tanh\left(\frac{\alpha_u^{(k)} - \mu_{k,e}}{\sigma_{k,e} + \varepsilon} \right), \tag{8}$$

where $\mu_{k,e}$ and $\sigma_{k,e}$ are computed over the finite values in the cutoff-safe calibration snapshot available at decision boundary e, and $\varepsilon > 0$ prevents division by zero. They are recomputed at the next cutoff; they are not estimated from the entry or target bars. The resulting signal $\tilde{\alpha}_u^{(k)} \in (-1, 1)$ is treated as a continuous directional score: positive values favour `LONG` and negative values favour `SHORT`.

3.4. Walk-Forward Alpha Selection

Let $\mathcal{A} = \{\alpha^{(1)}, \alpha^{(2)}, \ldots, \alpha^{(M)}\}$ denote the registry of $M = 85$ candidate alpha functions. Each point-in-time history ends at the decision cutoff and is therefore disjoint from the current entry and target bars. Given the net Open-to-Close execution-return series $r_u^{\text{net},(L)}$ defined as in Eq. (4), the following diagnostics are computed on finite historical observations:

The same matured calibration sample is used to estimate IC sign, orient the signal, calculate the three ranking diagnostics, and select the top five. Consequently, point-in-time truncation prevents future-data leakage but does not provide an independent inner/outer assessment of the adaptive selection rule. The equations below define the implemented selector; they should not be read as post-selection-unbiased estimates of an individual factor's skill.

Information Coefficient (IC).. Following Grinold and Kahn [29], the Spearman rank correlation between the normalised alpha signal and forward returns over the evaluation suffix:

$$\mathrm{IC}_k \;=\; \mathrm{SpearmanCorr}\big(\tilde{\alpha}^{(k)}_{\mathcal{T}_{\mathrm{eval}}},\; r^{\mathrm{net},(L)}_{\mathcal{T}_{\mathrm{eval}}}\big). \tag{9}$$

Directional Accuracy.. The fraction of bars where the predicted direction agrees with the realised direction, restricted to bars with a strong enough signal:

$$\mathrm{Acc}_k \;=\; \frac{1}{|\mathcal{T}^*|}\sum_{u\in\mathcal{T}^*}\mathbf{1}\big[\mathrm{sign}\big(\tilde{\alpha}^{(k)}_u\big) = y^{(L)}_u\big]\,, \tag{10}$$

where $\mathcal{T}^* = \big\{u : |\tilde{\alpha}^{(k)}_u| > 0.05\big\}$ excludes near-zero signals and $y^{(L)}_u$ is the canonical binary label from Eq. (4); hence a zero net return is `DOWN`, not a third class.

Long-Only Win Rate.. The precision of the alpha on bullish signals:

$$\mathrm{LongAcc}_k \;=\; \frac{\sum_u \mathbf{1}\Big[\tilde{\alpha}^{(k)}_u > 0.05 \;\wedge\; r^{\mathrm{net},(L)}_u > 0\Big]}{\sum_u \mathbf{1}\Big[\tilde{\alpha}^{(k)}_u > 0.05\Big]}. \tag{11}$$

Horizon Sharpe diagnostic.. For ranking, define $z^{(k)}_u = \mathrm{sign}(\tilde{\alpha}^{(k)}_u) r^{\mathrm{net},(L)}_u$ on the finite historical sample. Following Sharpe [30], the diagnostic is:

$$\mathrm{Sharpe}_k \;=\; \sqrt{\frac{252}{L}}\cdot\frac{\mathbb{E}_{u\in\mathcal{T}_{\mathrm{eval}}}\Big[z^{(k)}_u\Big]}{\mathrm{Std}_{u\in\mathcal{T}_{\mathrm{eval}}}\Big[z^{(k)}_u\Big] + \varepsilon}. \tag{12}$$

Sortino, maximum drawdown, hit rate, long-only accuracy, and average trade are retained as descriptive candidate diagnostics but do not enter the composite selection score.

Candidates with no fitted transform, fewer than 20 finite evaluation observations, or a non-finite value for any ranking metric are excluded before scaling. Let $\mathcal{V} \subseteq \mathcal{A}$ be the remaining eligible set and let $q \in \{|\mathrm{IC}|, \mathrm{Acc}, \mathrm{Sharpe}\}$. The implementation applies candidate-wise min–max scaling:

$$\mathcal{N}_{\mathcal{V}}(q_k) = \begin{cases} \dfrac{q_k - q_{\min}}{q_{\max} - q_{\min}}, & q_{\max} - q_{\min} \geq \varepsilon_{\mathrm{rank}}, \\ \dfrac{1}{2}, & \text{otherwise}, \end{cases} \tag{13}$$

where $q_{\min} = \min_{j\in\mathcal{V}} q_j$, $q_{\max} = \max_{j\in\mathcal{V}} q_j$, and $\varepsilon_{\mathrm{rank}} = 10^{-9}$. The default composite selection score is therefore:

$$S_k \;=\; 0.40\,\mathcal{N}_{\mathcal{V}}(|\mathrm{IC}_k|) \;+\; 0.35\,\mathcal{N}_{\mathcal{V}}(\mathrm{Acc}_k) \;+\; 0.25\,\mathcal{N}_{\mathcal{V}}(\mathrm{Sharpe}_k). \tag{14}$$

The weights prioritise association and directional accuracy while retaining a risk-adjusted return diagnostic. Long-only accuracy is reported but excluded from ranking to avoid favouring factors solely because the sample trends up.

Signal orientation is inferred from the sign of IC on the cutoff-safe historical snapshot. A negative-IC candidate is multiplied by -1 for its historical directional metrics and for the current inference value. All labels used for this operation have matured by the decision cutoff, but estimating orientation and ranking on the same finite history remains a selection risk discussed in Section 7.

Let $K = \min(5, |\mathcal{V}|)$. The top candidates are selected as:

$$\mathcal{A}^* \;=\; \operatorname*{arg\,max}_{\mathcal{B}\subseteq\mathcal{V},\,|\mathcal{B}|=K}\;\sum_{\alpha^{(k)}\in\mathcal{B}} S_k. \tag{15}$$

3.5. Sentiment Signal Construction

Let $\mathcal{D}_e = \{d_1, d_2, \ldots, d_{|\mathcal{D}_e|}\}$ be the set of CafeF articles available to the system for asset s at the timestamp τ_d of decision bar $d = e - 1$. Each article is assigned a score by the scorer actually recorded for that article:

$$\rho_i \;=\; \mathrm{Scorer}_i(d_i) \;\in\; [-1,\, 1], \tag{16}$$

where $\rho_i = +1$ denotes strongly positive, -1 strongly negative, and 0 neutral sentiment. For newly scored records, Scorer_i is either `5CD-AI/Vietnamese-Sentiment-visobert`, executed locally through Transformers or through a configured Hugging Face Dedicated Inference Endpoint, or the deterministic lexicon fallback. In automatic mode, a configured dedicated endpoint is attempted first and local inference is used if it fails; without an endpoint URL, local inference is selected directly. The classifier's official label mapping is $0 \mapsto$ NEG, $1 \mapsto$ POS, and $2 \mapsto$ NEU. Each record stores `scorer`, `scorer_backend`, and `is_fallback`. These fields identify the effective execution path but do not constitute an immutable model-revision digest. Legacy scores without these fields are marked `legacy-unknown`; no model identity is inferred from their numeric values.

The unweighted mean sentiment score for descriptive reporting is:

$$\bar{\rho}_e \;=\; \frac{1}{|\mathcal{D}_e|} \sum_{i=1}^{|\mathcal{D}_e|} \rho_i. \tag{17}$$

For the causal sentiment context, each dated article receives an exponential weight based on its age at cutoff τ_d:

$$\Delta_i(\tau_d) = \tau_d - \mathrm{date}(d_i), \tag{18}$$

$$w_i(\tau_d) = 2^{-\Delta_i(\tau_d)/h}, \qquad h = 30 \text{ days}, \tag{19}$$

$$S_{\mathrm{decay},e} = \frac{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d)\rho_i}{\sum_{d_i \in \mathcal{D}_e} w_i(\tau_d)}, \tag{20}$$

$$\mathrm{Rel}_{\mathrm{sent},e} \;=\; S_{\mathrm{decay},e} \;-\; \frac{1}{|\mathcal{R}_e|} \sum_{s' \in \mathcal{R}_e} S^{(s')}_{\mathrm{decay},e}, \tag{21}$$

where $\mathcal{R}_e$ contains only related companies with at least three valid dated-and-scored articles at cutoff τ_d. The exponential mean makes recent articles more influential without comparing a batch mean against the same batch from which it was calculated. $\mathrm{Rel}_{\mathrm{sent},e}$ compares like-for-like time-decayed scores.

Batch definition and low-news stability.. The set $\mathcal{D}_e$ contains all dated articles for target stock s in the 90-day sliding window ending at cutoff τ_d. The complete compact observation set is used for Eq. 20; the 15-article list stored for display is not the normalization sample. With fewer than three valid dated-and-scored observations, $S_{\mathrm{decay},e}$ is marked unavailable rather than assigned a neutral zero. Likewise, $\mathrm{Rel}_{\mathrm{sent},e}$ is unavailable when $\mathcal{R}_e$ is empty. The effective sample size, invalid-date/score counts, and number of post-cutoff observations excluded are stored in run provenance. Neither context variable enters any numerical alpha formula or the registry ranking.

To prevent look-ahead leakage during backtesting, every cached article stores timezone-aware `published_at` and `available_at` fields in `Asia/Ho_Chi_Minh`. The cache is keyed by (s, τ_d) and returns only articles satisfying $\mathrm{available_at}(d_i) \le \tau_d$. CafeF records having date-only precision are assigned a conservative availability time of 00:00 on the following local day; records without a parseable date are rejected in research mode and never used to fill a sparse historical window. All

180 articles in the nine supplied cache artifacts contain parseable dates, so the previous undated fallback was a latent risk rather than evidence that those reported runs actually consumed undated articles.

3.6. Multi-Agent Decision Function

The full system decomposes $\mathcal{F}$ into five sub-functions corresponding to the five specialised agents:

$$\mathbf{r}^{\text{ind}} = \mathcal{F}_{\text{ind}}\big(\mathbf{X}_e\big), \tag{22}$$

$$\mathbf{a}_e = \mathcal{F}_{\text{factor}}\big(\mathbf{X}_e,\ \mathcal{A}^*\big), \tag{23}$$

$$\mathbf{c}_e^{\text{sent}} = \mathcal{F}_{\text{context}}\big(\mathcal{D}_e,\ S_{\text{decay},e},\ \text{Rel}_{\text{sent},e}\big), \tag{24}$$

$$\mathbf{r}^{\text{alpha}} = \mathcal{F}_{\text{text}}\big(\mathbf{a}_e,\ \mathbf{c}_e^{\text{sent}}\big), \tag{25}$$

$$\mathbf{r}^{\text{pat}} = \mathcal{F}_{\text{pat}}\big(\mathbf{I}_e^{\text{kline}}\big), \tag{26}$$

$$\mathbf{r}^{\text{trend}} = \mathcal{F}_{\text{trend}}\big(\mathbf{I}_e^{\text{trend}}\big), \tag{27}$$

where $\mathbf{I}_e^{\text{kline}}$ and $\mathbf{I}_e^{\text{trend}}$ are the candlestick and trendline-annotated chart images rendered from $\mathbf{X}_e$, and $\mathbf{c}_e^{\text{sent}}$ is a separate textual context containing the target and related-company sentiment summaries. The factor values in $\mathbf{a}_e$ are functions of $\mathbf{X}_e$ only; the LLM compares them qualitatively with $\mathbf{c}_e^{\text{sent}}$ when producing $\mathbf{r}^{\text{alpha}}$.

The Decision Agent aggregates all four reports through structured chain-of-thought prompting [31] and produces the final binary signal:

$$\hat{y}_{e,L} = \mathcal{F}_{\text{dec}}\big(\mathbf{r}^{\text{ind}},\ \mathbf{r}^{\text{alpha}},\ \mathbf{r}^{\text{pat}},\ \mathbf{r}^{\text{trend}}\big) \in \{-1, +1\}. \tag{28}$$

The forced binary output constraint ensures $\hat{y}_{e,L} \neq 0$ for every valid *prediction*; it does not imply an always-invested account. Under the default long-only action map, `SHORT` becomes `CASH`. `LONG` executes a complete next-open-to-target-close cycle when valid positive prices and timestamps are available. This separation permits binary directional evaluation without misrepresenting a bearish prediction as an executed short sale.

3.7. Walk-Forward Evaluation Protocol

We adopt a rolling walk-forward evaluation protocol commonly used in quantitative trading system evaluation [32].

Given a historical series of T bars, valid test points are exclusive boundaries with $W \leq e \leq T - L$. The implementation uses the latest available boundaries at spacing S (up to N_{test} points) and processes them in chronological order:

$$\mathcal{T} = \big\{\, e_i = T - L - (i-1)\cdot S \ \mid\ i = 1, 2, \ldots, N_{\text{test}} \,\big\}, \tag{29}$$

retaining only values $e_i \geq W$ before reversing the list for execution. Thus $e = W$ is included whenever it is among the requested points, and every selected point satisfies the complete-window and complete-target conditions. Here S is the step size between consecutive test points and $N_{\text{test}} = 20$ in our main experiments.

The three primary evaluation metrics are:

$$\text{Acc} = \frac{1}{|\mathcal{T}|}\sum_{e\in\mathcal{T}}\mathbf{1}\big[\hat{y}_{e,L} = y_{e,L}\big], \tag{30}$$

$$\Delta\text{Acc} = \text{Acc}_{\text{Full}} - \text{Acc}_{\text{No-}\alpha}, \tag{31}$$

$$q_e = a_e\left[\frac{C_{e-1+L}}{O_e}\frac{(1-c_{\text{slip}})(1-c_{\text{fee}})}{(1+c_{\text{slip}})(1+c_{\text{fee}})} - 1\right], \tag{32}$$

$$W_{k+1} = W_k\,(1+q_{e_k})\,, \qquad W_0 = 1, \tag{33}$$

$$R_{\text{acct}} = \frac{W_K}{W_0} - 1, \tag{34}$$

where ΔAcc (*Alpha Lift*) quantifies the improvement in directional accuracy attributable to the Alpha Agent. In the economic simulation, $a_e = 1$ for `LONG` and $a_e = 0$ for `SHORT`. Each LONG allocates all available cash at O_e, pays fee and slippage on the BUY leg, liquidates all shares at C_{e-1+L}, and pays the same rates on the SELL leg. The configured values are $c_{\text{fee}} = 0.25\%$ and $c_{\text{slip}} = 0.10\%$ per leg. Each test point is a closed cycle and the implementation requires $S \geq L$, so no shares carry into the next point. SHORT forecasts hold cash and incur neither fee nor slippage. Let $\rho_k = W_{k+1}/W_k - 1 = q_{e_k}$ denote the cycle return, including zeros for CASH. The account risk metrics are

$$\text{Sharpe}_{\text{acct}} = \sqrt{252}\,\frac{\mathbb{E}[\rho_k]}{\text{Std}[\rho_k] + \varepsilon}, \tag{35}$$

$$\text{MDD}_{\text{acct}} = \max_{0\leq k\leq K}\left(1 - \frac{W_k}{\max_{0\leq i\leq k} W_i}\right). \tag{36}$$

The Sharpe risk-free rate is zero, and the population standard deviation is used. Maximum drawdown is a relative wealth loss and includes the initial wealth anchor W_0, so an initial losing cycle is not omitted.

Finally, let $\mathcal{X} = \{e_k : a_{e_k} = 1\}$ be the executed-position sample. Trade-level summaries are

$$\text{HitRate} = \frac{1}{|\mathcal{X}|}\sum_{e\in\mathcal{X}}\mathbf{1}[q_e > 0], \qquad \text{AvgTrade} = \frac{1}{|\mathcal{X}|}\sum_{e\in\mathcal{X}} q_e, \tag{37}$$

with both defined as zero when $\mathcal{X}$ is empty. SHORT/CASH cycles do not dilute either executed-trade denominator. Risk ratios are reported as zero when their dispersion denominator does not exceed the numerical tolerance. The cost rates are research allowances rather than a claim about a particular venue's realised fee schedule.

4. System architecture

4.1. Overview

The implemented runtime is a LangGraph directed acyclic graph with five specialised agents and a shared typed state object. It acts as an explicit knowledge-integration layer over technical, quantitative, textual, and visual evidence. Its default configuration names `openai/gpt-oss-20b` for text and `qwen/qwen3.8-27b` for vision through Groq. The backtest path enforces point-in-time input windows, shared upstream reports for paired variants, structured binary decisions, and a separate compounded-account ledger.

4.2. Indicator Agent

The Indicator Agent computes five momentum and volatility oscillators using TA-Lib over the most recent OHLCV window, including MACD [33], RSI [34], Stochastic Oscillator [35], Williams %R [36], and ROC-based momentum signals [37]. Table 1 lists each indicator with its parameters and signal classification rules. Signal classification is performed deterministically in Python before any LLM call, using hard thresholds and two-bar crossover detection. The classified counts-number of bullish, bearish, and neutral signals-are injected directly into the LLM prompt as immutable ground-truth constraints, preventing the model from hallucinating signal tallies. A majority rule then determines an overall consensus with an associated confidence level (High if four or more signals agree, Medium if three agree, Low otherwise).

Table 1: Technical indicators and trading signal conditions

Indicator	**Parameters**	**Bullish condition**	**Bearish condition**
MACD	12, 26, 9	Golden cross or positive histogram expanding	Death cross or negative histogram expanding
RSI	period = 14	≤ 30 (oversold) or value ≥ 55	≥ 70 (overbought) or value ≤ 45
Stochastic	14, 3, 3	%K crosses above %D, or %K ≤ 20	%K crosses below %D, or %K ≥ 80
Williams %R	period = 14	%R ≤ -80 or trending toward 0	%R ≥ -20 or trending toward -100
ROC	period = 10	ROC $> +0.5\%$	ROC $< -0.5\%$

4.3. Alpha Agent

The Alpha Agent serves two explicitly separated functions: OHLCV-only quantitative factor evaluation and qualitative comparison with Vietnamese sentiment supplied as textual LLM context.

Alpha selection.. An offline registry of exactly 85 candidate alpha functions is maintained, comprising five proprietary single-stock momentum and reversion formulas (Table 2) and 80 adaptations drawn from the WorldQuant 101 Formulaic Alphas. Of the original 101 WorldQuant expressions, only 80 are retained because the remaining 21 cannot be translated into a meaningful single-stock OHLCV setting without introducing information that is unavailable in our pipeline or fundamentally altering the original operator semantics. In particular, these excluded expressions rely on cross-sectional ranking or neutralisation across multiple contemporaneous assets, industry or market-group information, or other inputs/operators that do not admit a defensible time-series-only surrogate under our single-symbol formulation. We therefore exclude them rather than force an arbitrary approximation. Consequently, 101 denotes the size of the original source catalogue, whereas 80 denotes the subset considered adaptable to our single-stock setting, and 85 denotes the final executable candidate registry after adding five proprietary factors.

For the retained expressions, cross-sectional operators are replaced by single-asset time-series surrogates as formalised in Eq. (7). This translation is heuristic: without contemporaneous peer dispersion, a rolling percentile answers where is this stock relative to its own recent history?" rather than where is this stock relative to other stocks now?" The adaptation can therefore change factor scale, persistence, turnover, and economic meaning. The present experiments do not estimate per-factor IC-decay curves, lagged signal autocorrelation, or turnover profiles, and do not claim that

the adapted factors inherit the properties of their WorldQuant sources. When the volume series is unavailable or sparse, intra-bar price range is used as a proxy, creating an additional interpretation change for volume-dependent expressions.

For a given symbol and timeframe, the available point-in-time history is truncated to at most 600 bars ending at the decision cutoff. Features (returns, log-volume, VWAP proxy, and ADV20) are then constructed. Candidate metrics use only matured net Open-to-Close labels within this snapshot; the final L rows are excluded because their targets are unavailable at the cutoff. A zero net return is assigned `DOWN`, consistent with the requirement that a LONG must remain profitable after costs.

The selector computes Spearman Information Coefficient (IC), directional accuracy on signals above the configured absolute threshold, and a horizon Sharpe diagnostic. Long-only win rate and other economic summaries are retained for diagnosis but do not enter the composite score.

A candidate is eligible only when it has at least 20 finite observations and finite ranking metrics. Ineligible candidates are removed *before* the per-metric min–max bounds are computed. Using $\mathcal{N}_{\mathcal{V}}$ from Equation (13), the default composite score is:

$$S = 0.40\mathcal{N}_{\mathcal{V}}(|IC|) + 0.35\mathcal{N}_{\mathcal{V}}(Acc) + 0.25\mathcal{N}_{\mathcal{V}}(Sharpe) \tag{38}$$

If a metric is constant across the eligible set, its normalised value is 0.5 for every candidate. The displayed coefficients are the default runtime weights; robustness configurations replace them with recorded alternatives. Long-only accuracy is excluded from ranking to avoid rewarding a factor solely because the historical sample trends upward.

Up to five eligible candidates are selected for the current prediction. When historical IC is negative, the signal is multiplied by -1 for metric computation and current inference. All observations used for this orientation precede the decision cutoff, which prevents lookahead into the evaluated trade. It does not eliminate selection bias: the same trailing observations determine the IC sign, score all 85 candidates, and choose the five winners. A noisy negative IC can therefore be converted into an apparently useful positive orientation, and the maximum over many noisy scores is optimistic even when every timestamp is cutoff-safe. No nested inner-selection/outer-evaluation loop, Deflated Sharpe Ratio, or rolling inclusion/sign-stability diagnostic is applied in the reported study. Candidate IC and Sharpe values are consequently descriptive ranking statistics, not multiplicity-adjusted evidence that a particular factor is genuinely profitable. Nested validation that repeats orientation and top-five selection inside every training fold [17], together with Deflated Sharpe or false-discovery control [18, 15], is required for stronger factor-level claims.

Table 2: Five proprietary alpha formulas

Alpha	**Name**	**Formula**
α_1	Flow-Driven Momentum	$\tanh(\mathrm{ROC}(1)_{\mathrm{adj}} \times \mathrm{VolSurge} \times 0.5)$
α_2	MACD-Flow Proxy	$\tanh(\mathrm{MACDHist}_{\mathrm{norm}} + \mathrm{ROC}(1)_{\mathrm{adj}} \times 0.5)$
α_3	Liquidity Void Reversion	$\tanh\big(-0.8 \times \mathrm{ZScore}\big(\frac{\mathrm{Close}-\mathrm{SMA}_5}{\mathrm{Close}}\big)\big)$
α_4	Bollinger Squeeze & Flow	$\tanh((\%B - 0.5) \times \mathrm{VolZ}_5 \times 1.5)$
α_5	Order Flow Exhaustion	$\tanh(\mathrm{ROC}(1)_{\mathrm{adj}} \times \mathrm{ClosePos} \times 3.0)$

Sentiment integration.. Articles from CafeF are crawled for the target ticker and up to ten related companies identified by the LLM. The scorer supports four explicit backend policies: `auto`, `local`,

`endpoint`, and `lexicon`. The default `auto` policy calls a user-provisioned Hugging Face Dedicated Inference Endpoint when `HF_INFERENCE_ENDPOINT_URL` is configured and falls back to local Transformers inference if that call fails; without an endpoint URL, it selects local inference directly. The deprecated generic `hf-inference` serverless model route is not used. Local loading is lazy and thread-safe, uses the repository's `tokenizer.json` through `PreTrainedTokenizerFast`, truncates inference to 256 tokens, and selects CUDA when available unless the device is overridden. If the selected ViSoBERT path cannot produce a prediction, the versioned deterministic lexicon scorer is used. Every newly scored article stores the model/scorer name, the effective backend (local CPU/CUDA, dedicated endpoint host, lexicon, or empty-text handling), and a Boolean fallback flag.

Two point-in-time context variables are constructed. S_{decay} is the exponentially time-weighted mean of all valid dated-and-scored articles in the 90-day window, with a 30-day half-life and strict exclusion of post-cutoff observations. Rel_{sent} is the difference between the target's time-decayed score and the mean time-decayed score of related companies having at least three valid observations. Either value is marked unavailable when its observation requirement is not met; missing data is not encoded as neutral zero. The full compact observation set is used for these calculations, while the 15-article slice is display-only.

Neither variable is injected into α_2, any other registry formula, or factor ranking. They and the per-company related sentiment values are rendered in a separate prompt section that the LLM uses only for qualitative comparison with the OHLCV factor report.

In backtest mode, the live crawler is replaced by a pre-loaded historical sentiment store keyed by symbol and decision timestamp. Inclusion is based on timezone-aware `available_at`, not merely a publication date: date-only CafeF records become available at 00:00 on the next day in `Asia/Ho_Chi_Minh`, and undated records are rejected rather than used as fallback.

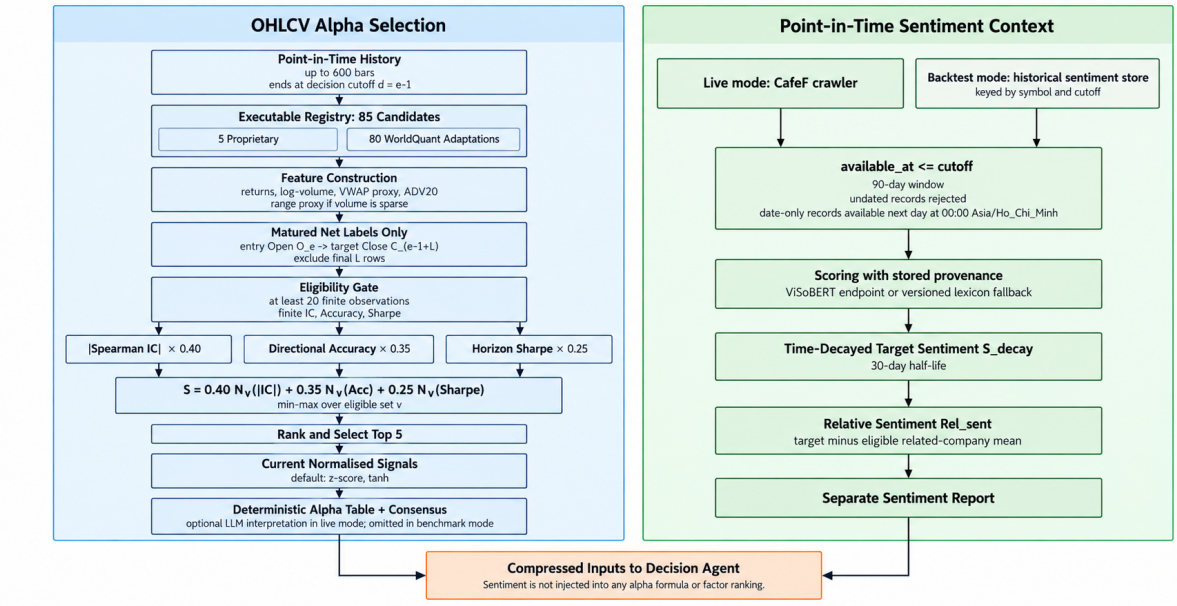


Figure 1: Internal data flow of the Alpha Agent

4.4. Pattern Agent

The Pattern Agent analyses a candlestick chart rendered from the current analysis window $[e - W, e)$ (45 candles in the main configuration). The chart is produced by `mplfinance` using a

custom colour scheme (green bodies for bearish candles and red bodies for bullish candles, following Vietnamese market convention) and encoded as a base64 PNG before graph execution.

The vision-language model is prompted to perform four sequential tasks. First, it identifies the dominant chart formation from a curated list of 16 classical patterns, such as head-and-shoulders inverse, double bottom, or various triangles and wedges. Second, it describes the three to five most recent candles in terms of body size, wick length, and relative position. Third, it assesses whether a breakout or breakdown is occurring. Finally, it commits to a directional bias (`bullish`, `bearish`, or `neutral`) alongside a confidence rating of `High`, `Medium`, or `Low`.

The prompt is explicitly structured to discourage non-committal answers and encourage a directional interpretation whenever sufficient evidence is present.

After inference, a deterministic verifier compares the reported bias with the final candlestick body and the direction of the last five closes. An obvious contradiction is appended to the report and downgrades its confidence. If the vision model fails or no chart image is available, the agent falls back to a text-only analysis of raw OHLCV values using the text LLM.

4.5. Trend Agent

The Trend Agent receives a separate chart in which support and resistance lines have been fitted algorithmically before LLM invocation.

Two trendline pairs are computed.

Close-price trend channel.. The first pair is derived from the closing price series alone. A linear regression line is fitted to all 45 candles in $[e - W, e)$, and the upper and lower deviations from this line identify the resistance and support pivots respectively. The slopes are then refined by minimising squared deviations subject to two constraints: the support line must not intersect any closing price from above, and the resistance line must not intersect any closing price from below.

High-low channel.. The second pair uses the high and low price series to identify the highest and lowest deviation points, producing channel lines that bound the full intra-bar range.

The support lines are drawn in blue, resistance lines in red, and rendered at 600 DPI to preserve legibility when compressed to base64.

The vision-language model is prompted to assess the slope of each support and resistance line, evaluate the distance between the lines (indicating channel width and price compression), determine the position of the most recent candles relative to each boundary, and produce a directional forecast for the applicable prediction horizon. The required JSON output fields encompass the trend direction, numerical support and resistance levels, slope characterisation, and overall confidence.

After inference, a deterministic verifier checks the reported trend against price relative to SMA20, the availability of SMA50, and the recent price slope. Contradictory claims are flagged and their confidence is reduced. If the vision-language model fails or no rendered chart is available, the agent falls back to a text-only analysis of raw OHLCV data, mirroring the strategy used by the Pattern Agent.

Scope of visual verification.. The two deterministic checks are *output-consistency gates*: they test whether the final directional statement contradicts a small set of OHLCV-derived facts. They cannot reveal which image region caused the model's answer, verify every claimed candle or trendline coordinate, or demonstrate sensitivity to the purported local pattern. A model that simply extrapolates the global trend can pass both gates whenever the shortcut and the rules agree. This distinction matters because observational financial charts couple local evidence with trend, making ordinary accuracy unable to identify grounding; controlled evidence interventions are needed to separate a

grounded responder from a trend-shortcut responder [11]. Accordingly, the evidence establishes reduced obvious contradiction, while audited visual faithfulness remains an open question. A direct audit would require matched charts that preserve global trend while altering only local candlestick semantics, complementary trend swaps that preserve the local pattern, and measurement of whether the model's direction changes in the rule-implied way. FinChart-Bench further cautions that current models remain unreliable on spatial and instruction-following aspects of financial charts [10].

4.6. Decision Agent

The Decision Agent synthesises all upstream reports into a single trading judgment. To remain within the Groq token-per-minute limit, each report is condensed before being passed to the prompt: the Indicator report is trimmed to its consensus summary block, the Alpha report retains only the factor table and LLM reasoning paragraph, and the Sentiment report drops the per-article listing.

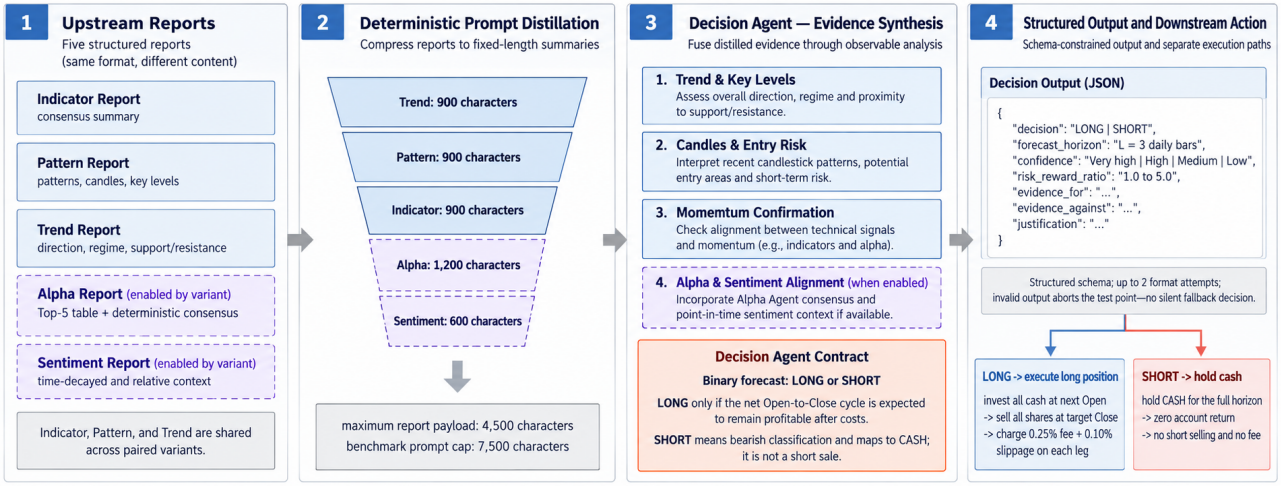


Figure 2: Decision Agent reasoning pipeline

The agent is prompted to reason through four sequential questions in a structured chain-of-thought process. It evaluates whether the price is near support or resistance while identifying the primary trend, checks if recent candles show reversal or continuation signals to ensure a low-risk entry, verifies whether technical oscillators like RSI and MACD confirm the price direction, and finally assesses if the quantitative alpha signals and sentiment align with or contradict the technical picture.

Following this reasoning, the model must output a JSON object comprising a forced binary direction forecast (`LONG` or `SHORT`), the `forecast_horizon`, the `confidence` level, a `risk_reward_ratio` clamped to $[1.0, 5.0]$, supporting and opposing evidence, and a final `justification` paragraph. A separate prompt instruction distinguishes this forecast label from an executed order.

The forced binary constraint supports direction classification only. Downstream account logic independently maps predictions to actions: `SHORT` maps to `CASH`, whereas `LONG` invests all available cash at the next open and closes the position at the configured target close after applying costs on both legs.

5. Experimental Setup

5.1. Dataset and Point-in-Time Data Contract

The daily evaluation universe contains nine Vietnamese equities spanning five economically distinct groups: information technology (FPT and CMG), banking (VCB and MBB), retail (MWG), food and beverages (VNM and BHN), and aviation and tourism (HVN and VJC). The purposive stratification combines the VN30/large-cap names specified in the frozen design (VCB, FPT, VNM, MBB, and MWG) with mid-cap or ownership-specialised firms (CMG, BHN, and HVN) and a large private airline (VJC). It was chosen to expose the agents to heterogeneous liquidity, ownership, and news environments; it is not claimed to be a random or market-representative sample. The benchmark contains 20 chronologically ordered test points per symbol, for 180 paired observations in total. Section 7 discusses the resulting external-validity limit, and Appendix Table E.16 reports the complete ticker and sentiment-coverage audit.

OHLCV observations are obtained through the project's `vnstock`-based data loader. At an exclusive test boundary e, every agent sees only bars in $[e - W, e)$, so the last visible decision bar is $d = e - 1$. Dynamic alpha selection receives a cutoff-safe prefix ending at the same decision bar and containing at most 600 observations. Neither the factor calculations nor the chart renderers can inspect the entry bar or target bar. The numerical, candlestick, and trend inputs use the same requested window length and fail closed when a complete window is unavailable.

The historical sentiment store applies the same cutoff. Only records with a parseable publication date no later than d are eligible; undated records are rejected in research mode instead of being borrowed from a later cache state. Across the supplied caches, all 180 articles have a parseable date. The mean number available in a 90-day point-in-time window is 5.77, and 73 of 180 test points (40.56%) contain at least eight articles. The legacy cache schema does not retain the scorer identity or fallback flag, so 0 of 180 articles can be verified as ViSoBERT-scored, 0 can be verified as lexicon-scored, and all 180 are conservatively reported as `legacy-unknown`. These figures measure verifiable provenance rather than infer model identity from confidence or sentiment values. The sentiment pathway remains text context for the Decision Agent and does not enter any OHLCV alpha formula.

5.2. Walk-Forward Protocol

All reported experiments use daily observations. The main configuration is $W = 45$, step size $S = 3$, and lookahead $L = 3$. For boundary e, the simulated entry is the next open O_e and the target exit is C_{e-1+L}. The realised net return of a candidate LONG cycle is

$$r_{e,L}^{\text{net}} = \frac{C_{e-1+L}}{O_e} \frac{(1 - c_{\text{slip}})(1 - c_{\text{fee}})}{(1 + c_{\text{slip}})(1 + c_{\text{fee}})} - 1, \tag{39}$$

where $c_{\text{fee}} = 0.0025$ and $c_{\text{slip}} = 0.001$ are applied separately on both the BUY and SELL legs. The economic direction label is `UP` only when $r_{e,L}^{\text{net}} > 0$ and is `DOWN` otherwise. This definition makes the classification target, alpha-selection target, and account return share the same Open-to-Close horizon and cost convention. Figure 3 summarizes the timing contract, per-cutoff processing sequence, paired shared-report comparison, and separate account ledgers.

5.3. Paired Baseline and Disentangled Ablations

The legacy benchmark compares Full with a control historically named No-Alpha, here called Base reports. The control removes both OHLCV factors and sentiment. The implemented paired shared-report protocol intends to deep-copy Indicator, Pattern and Trend reports within each pair; complete historical reports and requests have not been recovered to independently certify every

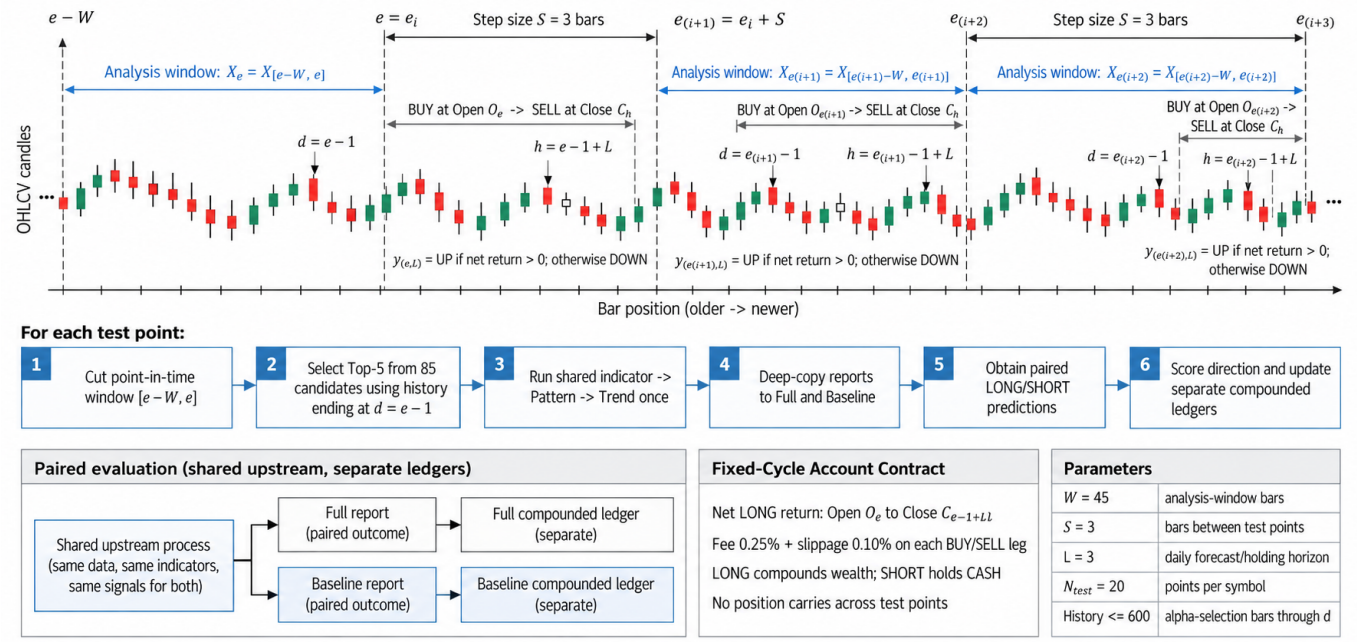


Figure 3: Walk-forward evaluation protocol.

Table 3: Main walk-forward configuration.

Parameter	**Value**	**Operational meaning**
Analysis window W	45 bars	Exact agent input $[e-W, e)$
Step size S	3 bars	Separation between consecutive test boundaries
Lookahead L	3 bars	Entry at O_e and exit at C_{e-1+L}
Test points	20 per symbol	180 paired observations over nine symbols
Alpha history	Up to 600 bars	Cutoff-safe calibration/evaluation prefix ending at $d = e-1$
Alpha registry	85 candidates	Top five selected at each test point
Random seeds	42	Python/NumPy and bootstrap reproducibility

archived pair. All 180 saved pairs contain valid binary decisions. Their accuracy difference is a Full–Base bundle gap, not an estimate isolating alpha information.

The legacy four-way experiment on FPT, VNM and VCB is intended to compare Full, Alpha-only, Sentiment-only and Base reports with shared upstream inputs. Its branch provenance remains unresolved, so its stored outcomes do not establish independent component contributions. The historical summary below measures improvement over the best single component, not a factorial interaction:

$$\text{Synergy} = \text{Acc}_{\text{Full}} - \max(\text{Acc}_{\text{AlphaOnly}}, \text{Acc}_{\text{SentimentOnly}}) . \tag{40}$$

5.4. Account and Statistical Metrics

Each variant starts with VND 50,000,000 and is simulated independently. Every LONG uses all available cash at O_e, pays fee and slippage on purchase, sells all shares at C_{e-1+L}, and pays both costs again on sale. Every cycle therefore ends in cash before the next test point. A SHORT forecast

Table 4: Paired benchmark variants.

Component	**Full System**	**No-Alpha baseline**
Indicator report	Shared upstream snapshot	Identical shared snapshot
Pattern and Trend reports	Shared upstream snapshot	Identical shared snapshot
OHLCV alpha factors	Enabled	Disabled
Vietnamese sentiment context	Enabled	Disabled
Final Decision Agent	Receives all enabled reports	Receives only shared upstream reports

is a directional prediction only and maps to CASH because uncovered short selling of the underlying equity is disabled. With cycle return $q_k = r^{\text{net}}_{e_k,L}$ for LONG and $q_k = 0$ for SHORT, wealth compounds as

$$W_{k+1} = W_k(1 + q_k), \qquad R_{\text{acct}} = \frac{W_K}{W_0} - 1. \tag{41}$$

The legacy files store Sharpe calculated with $\sqrt{252}$ on cycle returns; this frequency convention is under methodological correction and is not used here to support an investment claim. Their drawdown is cycle-end rather than daily mark-to-market. The new numerical ledger additionally marks holdings at every observed session close, records daily quoted-notional turnover divided by prior equity, and separates held-session fraction from close-of-day exposure. Continuous holding buys once at the first entry and sells once at the final exit, including inter-cycle gaps. It is a price-series comparator, not a verified total-return account: corporate-action/cash-dividend ledgers are unavailable. Fractional shares, immediately reusable cash, zero interest, and the stated two-sided costs remain idealised research assumptions, not a venue-calibrated execution model.

Legacy McNemar, bootstrap, Newey–West and cross-symbol Wilcoxon outputs are retained as historical diagnostics with their original assumptions. They are not used to assert general significance, since 20 observations per symbol and common market dates do not constitute nine independent replications. The numerical reconstruction is descriptive and reports no superiority test. Balanced accuracy is the primary predictive summary, with MCC, macro-F1, raw accuracy, class support and both classes' precision/recall/F1 also reported. Undefined denominators are NA rather than zero. Pooled metrics aggregate confusion counts; macro-over-symbol metrics weight each symbol equally and require each symbol's metric to be defined.

5.5. Exploratory Numerical Baselines and Comparability

The additional numerical evaluation freezes a newly retrieved VCI price series through 30 September 2026 and retains all 20 legacy cutoff dates for each symbol. This is a reconstruction, not an authenticated copy of the historical LLM input snapshot. Canonical entry/exit prices and net labels are compared before any Full comparison is permitted. The 74 mismatched endpoints and two changed labels prevent that comparison, even though pooled class counts happen to remain 56 UP and 124 DOWN.

Training decisions end in 2025; validation begins in January 2026 and ends before each symbol's first test decision. Every training label exits strictly before the first validation decision, and every validation label exits strictly before the first test decision. Samples use three-session spacing and 300 warm-up sessions. An invalid BHN OHLC row on 13 September 2022 has no verified replacement, so the numerical feature/training view starts at the valid suffix after that row; the raw file and all test dates remain unchanged.

Table 5: Evaluation metrics and estimands.

Metric	**Definition**
Balanced accuracy	Mean of UP and DOWN recalls; both class supports must be nonzero.
MCC and class metrics	Matthews correlation; macro-F1; class supports and precision/recall/F1, with undefined values reported as NA.
Directional accuracy	Fraction of valid LONG/SHORT forecasts matching the net economic direction in Eq. (4).
Legacy bundle gap	Full accuracy minus Base reports accuracy; not isolated alpha information value.
Account return	Compounded fixed-cycle return $W_K/W_0 - 1$; LONG executes and SHORT holds cash.
Account Sharpe	Mean cycle return divided by population standard deviation and annualised by $\sqrt{252}$.
Maximum drawdown	Largest peak-to-trough decline on the compounded cycle-end equity curve.
Paired inference	Exact McNemar test, moving-block 95% lift interval, Newey–West mean test, and cross-symbol Wilcoxon signed-rank test.

Baselines are constant DOWN/CASH, always-LONG cycles, train-fitted majority, fixed 20-session momentum, logistic regression on price/indicator features, logistic regression on the same price features plus 85 factors and verified sentiment/missingness, and equal factor voting with train-only orientation. Five factor formulas containing whole-prefix normalisation receive explicit expanding past-only adapters in the new baseline pipeline; the legacy registry is unchanged. The broader factor set, absent peer context and missing reports mean that the second logistic is an information-family comparator, not a verified reconstruction of Full's exact numerical/textual inputs. Unknown historical scorer provenance makes sentiment unavailable rather than neutral.

Median imputation, missingness indicators and standardisation are fitted only on train during selection. The fixed regularisation grid is $C \in \{0.01, 0.1, 1, 10\}$ and threshold grid is $\{0.5, 0.4, 0.6\}$, selected by validation balanced accuracy with fixed-order ties. Selected preprocessing and logistic models are then refitted on train plus matured validation. A seeded random control uses the training momentum LONG rate; it is explicitly not a Full-exposure-matched control, since historical Full train predictions are missing. Causal per-symbol regimes use the close/SMA60 gap (above 2%, below −2%, or otherwise sideways). One deterministic numerical run is reported; it is not substituted for repeated LLM runs or multi-year LLM folds.

5.6. Implementation and Robustness Design

The text model is `openai/gpt-oss-20b` and the vision model is `qwen/qwen3.8-27b`; both use temperature zero in the reported runs. Text and vision output caps are 2,048 and 1,024 tokens, respectively, with at most three client retries and hidden reasoning format requested for supported reasoning models. Python and NumPy seed 42 controls local experiment ordering and deterministic utilities; the moving-block bootstrap uses seeds 42, 43, and 44 for Full, No-Alpha, and paired-lift resamples. No seed is passed to the hosted decoding API. Temperature zero therefore reduces sampling variation but does not imply bitwise determinism.

The configured provider identifiers are recorded, but the Groq endpoint did not expose an immutable weight digest or provider-side revision in the archived run artifacts, and the aliases were

not pinned to such a revision. Exact replay across a later provider update is therefore not guaranteed. This limitation is separate from checkpoint reproducibility: calls are executed sequentially with rate-limit pacing, and resuming a checkpoint preserves completed outputs instead of regenerating them. The legacy sentiment caches have an analogous, explicitly reported provenance limit: their article scores predate scorer metadata and are labelled `legacy-unknown` rather than attributed retrospectively to a ViSoBERT revision. The corrected runtime records `scorer`, `scorer_backend`, and `is_fallback` for newly scored articles, but this software change neither identifies the backend of historical cache entries nor changes any reported benchmark value.

Table 6: Implementation configuration used by the reported experiments.

Component	**Configuration**
Text / vision models	`openai/gpt-oss-20b` / `qwen/qwen3.8-27b`; temperature 0.0
Decoding controls	Text/vision caps 2,048/1,024 tokens; three client retries; hidden reasoning requested where supported
Randomness scope	Python/NumPy seed 42; bootstrap seeds 42–44; no hosted-decoding seed
Model revision	Provider aliases recorded; immutable Groq weight revision/digest unavailable and not pinned
Initial capital	VND 50,000,000 independently for each variant
Costs	0.25% fee and 0.10% slippage on each BUY and SELL leg
Position rule	LONG invests all cash for one fixed cycle; SHORT remains in CASH
Primary protocol	`paired_shared_reports`; 20 test points per symbol
Four-way protocol	`four_way_shared_upstream`; Full, Alpha-only, Sentiment-only, Baseline

The robustness experiment evaluates FPT and VNM with 20 test points per configuration. Panel A changes the common analysis window to $W \in \{30, 45, 60\}$. Panel B compares `zscore_tanh`, `minmax`, and `rank` alpha normalisation. Panel C compares default, equal, and IC-heavy ranking weights. For Panels B and C, the No-Alpha branch is reused from the same shared upstream/control checkpoint within each symbol and panel; its decisions must therefore be invariant to parameters that only affect the Alpha branch. Panel A deliberately changes the shared input window, so its No-Alpha output is not expected to remain constant across rows.

6. Backtest Results

6.1. Nine-Symbol Paired Benchmark

The original 180 net labels contain 56 UP and 124 DOWN outcomes. Full has confusion counts TP/FN/TN/FP of 16/40/101/23, compared with 15/41/77/47 for Base reports. Its 65.0% accuracy is below constant DOWN/CASH at 68.9%. Full's balanced accuracy is 55.01% and MCC is 0.113; the Base reports values are 44.44% and −0.108. Full issues 39 LONG calls versus 62 for Base reports. Of 25 additional correct Full decisions, 24 are DOWN and one is UP. This pattern is consistent with fewer LONG decisions in a predominantly unprofitable-cycle sample, but does not identify the cause as alpha information. Tables 7 and 8 report the imbalance-aware audit without suppressing the stronger constant baseline.

Table 9 retains the historical paired summaries for traceability. Its Full–Base gap is a bundle comparison. Stored cross-symbol Wilcoxon and small-sample intervals are descriptive legacy diagnostics, not evidence of a general significant alpha or forecasting improvement.

Table 7: Legacy class-imbalance audit on the original 180 labels and cycle endpoints.

Variant	N	Acc. (%)	BA (%)	MCC	Macro-F1 (%)	LONG (%)	Mean return (%)
Full (legacy)	180	65.00	55.01	0.113	54.96	21.67	-1.68
Base reports (legacy)	180	51.11	44.44	-0.108	44.53	34.44	-8.43
Constant DOWN / CASH	180	68.89	50.00	NA	40.79	0.00	0.00
Always-LONG cycles	180	31.11	50.00	NA	23.73	100.00	-17.69

Predictive metrics are pooled; return is the mean of nine independent accounts, not a pooled portfolio. UP/DOWN support is 56/124. Undefined values are NA. Always-LONG closes and reopens each cycle; it is not buy-and-hold.

Table 8: Per-class legacy precision, recall and F1, in percent.

Variant	Class	Support	Precision	Recall	F1
Full (legacy)	UP	56	41.03	28.57	33.68
Full (legacy)	DOWN	124	71.63	81.45	76.23
Base reports (legacy)	UP	56	24.19	26.79	25.42
Base reports (legacy)	DOWN	124	65.25	62.10	63.64
Constant DOWN / CASH	UP	56	NA	0.00	0.00
Constant DOWN / CASH	DOWN	124	68.89	100.00	81.58
Always-LONG cycles	UP	56	31.11	100.00	47.46
Always-LONG cycles	DOWN	124	NA	0.00	0.00

A class with support but no predicted positives has undefined precision and zero recall/F1. All symbol-level confusion matrices and class metrics are retained in the reanalysis CSV/JSON.

Full's recomputed mean account return is -1.6833%; averaging the already rounded legacy percentages instead gives -1.6856%, explaining the historical -1.69% display. Both are below CASH at zero. These are means of nine independent accounts, not a joint portfolio. Correct SHORT predictions earn no return under the long-only map. Neither the Full–Base accuracy gap nor relative loss reduction establishes isolated alpha value or profitability.

6.2. Numerical Reconstruction and Daily Ledgers

Tables 10 and 11 report the additional numerical study under Section 5.5. These results belong to the new snapshot and are not ranked against Full's legacy outcomes. The price-only and factor-augmented logistic models answer different information-set questions; neither has verified identical archived Full reports. The continuous-hold account buys once and sells once, unlike always-LONG cycles. Daily-close drawdown, turnover, fees and exposure are retained per symbol in the reanalysis ledgers; no daily Full ledger is inferred from mismatched reconstructed prices. The train-rate random control is not represented as matching Full's exposure. Accordingly, the newly implemented baselines do not by themselves close the common-snapshot and same-information requirements of a confirmatory baseline comparison.

6.3. Robustness and Sensitivity

Table 12 reports all six completed sweeps. Each row contains 20 paired test points. Panel A changes the input window for both variants and therefore changes the No-Alpha information set. In Panels B and C, which intervene only on Alpha processing, the No-Alpha accuracy is exactly invariant within each symbol and panel: 45.0% for both normalisation sweeps, 50.0% for the FPT weight sweep, and 45.0% for the VNM weight sweep.

Table 9: Paired benchmark over nine equities ($N = 20$ per symbol, $W = 45$, $S = 3$, $L = 3$). Panel A reports directional inference; Panel B reports compounded long-only account outcomes.

Panel A: Directional accuracy and paired inference

Symbol	**Full (%)**	**No-α (%)**	**Lift (pp)**	**95% CI (pp)**	**McNemar p**	N
BHN	55.0	45.0	+10.0	[+0.0, +15.0]	0.6875	20
CMG	75.0	70.0	+5.0	[+0.0, +5.0]	1.0000	20
FPT	65.0	45.0	+20.0	[−5.0, +35.0]	0.2188	20
HVN	70.0	50.0	+20.0	[+0.0, +35.0]	0.2188	20
MBB	70.0	45.0	+25.0	[+0.0, +55.0]	0.1250	20
MWG	70.0	50.0	+20.0	[+0.0, +40.0]	0.1250	20
VCB	65.0	50.0	+15.0	[+5.0, +35.0]	0.2500	20
VJC	60.0	60.0	+0.0	[−10.0, +10.0]	1.0000	20
VNM	55.0	45.0	+10.0	[−15.0, +45.0]	0.6250	20
Mean / total N	**65.0**	**51.1**	**+13.9**	–	–	**180**

Panel B: Compounded account metrics

Symbol	**Return F (%)**	**Return N (%)**	**Sharpe F**	**Sharpe N**	**MDD F (%)**	**MDD N (%)**
BHN	-4.20	-5.76	-3.63	-2.64	4.80	8.31
CMG	-2.66	-5.60	-3.64	-5.28	2.66	5.60
FPT	-1.56	-10.37	-0.86	-6.41	6.30	10.37
HVN	+2.89	-15.64	+1.19	-4.27	5.27	20.06
MBB	+5.21	-7.78	+2.36	-4.22	3.50	9.89
MWG	+0.39	-6.42	+0.41	-3.93	1.92	8.17
VCB	-8.34	-10.49	-5.98	-7.78	8.92	10.49
VJC	-6.05	-6.26	-5.22	-5.38	6.05	6.26
VNM	-0.85	-7.51	-0.41	-4.91	3.40	7.51
Mean	**-1.69**	**-8.43**	**-1.75**	**-4.98**	**4.76**	**9.63**

F and N denote Full and No-Alpha. The 95% intervals use a moving-block bootstrap (block length 5; 1,000 replications). The two-sided cross-symbol Wilcoxon result is $p = 0.0078125$ (nine symbol-level lifts; median lift = 15.0 pp). Returns are compounded from VND 50,000,000 independent accounts under Eq. (34); SHORT forecasts remain in cash.

The stored alpha-enabled branch varies with configuration. These sweeps do not validate historical branch provenance or establish alpha information value: snapshot dates differ between experiments, repeats are absent, and the control may be weak. Their arithmetic is retained for traceability, while substantive robustness conclusions await a verified common-snapshot study.

6.4. Disentangled Alpha–Sentiment Ablation

Table 13 retains the legacy four-way summaries for traceability only. Identical prediction sequences across differently labelled FPT branches and missing request archives leave provenance unresolved. The numerical gaps in this table therefore cannot be called independent alpha or sentiment contributions. Its final column is the stored improvement over the best single component, not the factorial interaction.

No component-information conclusion is drawn from these unverified summaries. Their presence

Table 10: Exploratory numerical baselines on the newly retrieved snapshot: not a head-to-head comparison with legacy Full.

Variant	Acc. (%)	BA (%)	MCC	Mean return (%)	Mean daily MDD (%)	Held sessions (%)
Constant DOWN / CASH	68.89	50.00	NA	0.00	0.00	0.00
Always-LONG cycles	31.11	50.00	NA	-17.33	21.13	99.15
Continuous hold (price only)	NA	NA	NA	-5.53	14.74	100.00
Factor-only vote	61.67	62.38	0.230	-5.07	11.93	46.75
Logistic: price	66.11	52.88	0.077	-1.15	4.22	13.80
Logistic: price + factors + S/M	61.67	51.12	0.025	-2.83	7.38	21.62
Momentum (20)	50.00	46.08	-0.074	-8.55	10.50	40.60
Train majority	68.89	50.00	NA	0.00	0.00	0.00
Train-rate random control	50.56	47.47	-0.047	-10.03	12.06	42.44

All cycle classifiers use the same 180 new-snapshot cutoffs and net targets. Predictive metrics are pooled; economic columns average independent symbol accounts. S/M denotes verified sentiment and missingness; historical scorer provenance is unavailable, so sentiment scores are missing. The train-rate control uses training momentum exposure, not Full exposure. Continuous hold has one buy and one sell per symbol; price adjustments and dividends are not independently certified. No Full row is included because 74 cycles differ at entry and/or exit from legacy, including two changed labels.

Table 11: Training support and snapshot comparability audit.

Symbol	Train	Validation	Test	Different endpoints	Label changes
BHN	175	34	20	20	2
CMG	399	36	20	1	0
FPT	399	36	20	20	0
HVN	399	36	20	1	0
MBB	399	36	20	1	0
MWG	399	36	20	20	0
VCB	399	36	20	9	0
VJC	399	36	20	0	0
VNM	399	36	20	2	0

Each label exit precedes the next split's first decision. BHN uses a valid suffix beginning 14 September 2022 after an unverified OHLC contradiction; raw data are preserved. All 20 test dates are retained for every symbol. Matching aggregate class counts does not establish matching labels or snapshots.

preserves the audit trail; replacing them as confirmatory evidence requires verified factorial/placebo branches on common data and repeated runs, rather than interpreting arithmetic differences as information value.

7. Limitations & Future Work

7.1. Limitations

L1. Statistical scale and economic interpretation.. The legacy benchmark contains only 20 observations per symbol. Common dates and market shocks invalidate interpreting nine symbol-level gaps as nine independent market replications, and the stored Wilcoxon result is not used to establish a general significant improvement. The tickers are not a probability sample. Class imbalance, snapshot revisions and incomplete input provenance further constrain inference even for this fixed universe.

Directional skill does not directly imply profitability. The Full System improves mean accuracy by 13.9 pp and outperforms No-Alpha in account return for every symbol, yet its mean compounded

Table 12: Robustness sweep on FPT and VNM ($N = 20$ per row). Returns are compounded account returns.

Symbol	Panel	Scenario	Full (%)	No-α (%)	Lift (pp)	Return F (%)	Return N (%)
FPT	Window	$W = 30$	55.0	40.0	+15.0	-2.63	-10.27
FPT	Window	$W = 45$	65.0	50.0	+15.0	-2.21	-6.86
FPT	Window	$W = 60$	40.0	45.0	-5.0	-8.04	-4.38
VNM	Window	$W = 30$	40.0	40.0	+0.0	-7.81	-7.51
VNM	Window	$W = 45$	55.0	40.0	+15.0	-3.59	-7.06
VNM	Window	$W = 60$	65.0	55.0	+10.0	+5.20	-0.84
FPT	Normalisation	`zscore_tanh`	50.0	45.0	+5.0	-4.72	-7.38
FPT	Normalisation	`minmax`	55.0	45.0	+10.0	-5.63	-7.38
FPT	Normalisation	`rank`	60.0	45.0	+15.0	-4.36	-7.38
VNM	Normalisation	`zscore_tanh`	60.0	45.0	+15.0	+2.70	-2.22
VNM	Normalisation	`minmax`	65.0	45.0	+20.0	+3.89	-2.22
VNM	Normalisation	`rank`	55.0	45.0	+10.0	-2.01	-2.22
FPT	Weighting	`default`	60.0	50.0	+10.0	-1.99	-7.85
FPT	Weighting	`equal_weight`	60.0	50.0	+10.0	-0.36	-7.85
FPT	Weighting	`ic_heavy`	65.0	50.0	+15.0	-2.21	-7.85
VNM	Weighting	`default`	65.0	45.0	+20.0	+6.17	-5.19
VNM	Weighting	`equal_weight`	50.0	45.0	+5.0	-4.57	-5.19
VNM	Weighting	`ic_heavy`	60.0	45.0	+15.0	+2.70	-5.19

Table 13: Unverified legacy four-way summaries ($N = 20$ per variant and symbol); not confirmatory attribution evidence.

Panel A: Directional decomposition

Symbol	Full	Alpha-only	Sent.-only	Baseline	$\Delta\alpha$	ΔSent.	Synergy
FPT	55.0%	50.0%	45.0%	45.0%	+5.0 pp	+0.0 pp	+5.0 pp
VNM	60.0%	50.0%	50.0%	45.0%	+5.0 pp	+5.0 pp	+10.0 pp
VCB	45.0%	55.0%	40.0%	40.0%	+15.0 pp	+0.0 pp	-10.0 pp
Mean	**53.33%**	**51.67%**	**45.00%**	**43.33%**	**+8.33 pp**	**+1.67 pp**	**+1.67 pp**

Panel B: Compounded account return

Symbol	Full	Alpha-only	Sentiment-only	Baseline
FPT	-4.61%	-7.85%	-7.38%	-6.09%
VNM	-4.03%	+2.26%	-5.86%	-3.96%
VCB	-7.93%	-2.75%	-9.23%	-11.34%

return remains −1.69%, with only three symbols positive. Because SHORT predictions map to CASH, correct bearish forecasts generate no trading return. The observed directional lift therefore does not establish deployable profitability.

L2. Simplified execution model.. The fixed-cycle ledger aligns labels, alpha targets, and account returns on the same next-open-to-target-close net return and includes two-leg fees and slippage. However, these costs remain benchmark assumptions. The simulator omits dynamic spreads, taxes, order-book depth, partial fills, market impact, position limits, and cross-symbol capital competition. It also assumes each $S = L = 3$ cycle is closed before the next entry. A venue-calibrated event-driven

simulator is therefore required before live deployment.

L3. Alpha selection and sensitivity.. Selecting five factors from 85 candidates introduces multiple-testing and winner's-curse risk. Although point-in-time cutoffs prevent future-bar leakage, the same trailing sample is used to estimate IC sign, compute oriented Accuracy and Sharpe, and rank the factors. Selecting large composite scores across many candidates can therefore produce optimistic diagnostics. No nested cross-validation, Deflated Sharpe Ratio, multiplicity correction, or sign-stability criterion was applied. The benchmark evaluates the complete adaptive policy at later cutoffs, but its per-factor statistics should not be interpreted as unbiased factor discoveries.

The 80 WorldQuant-derived entries are heuristic single-asset adaptations rather than exact implementations of the original cross-sectional formulas. Their stability was not evaluated through IC decay, lagged signal autocorrelation, or turnover. These diagnostics are needed to determine whether apparent predictiveness survives to the $L = 3$ horizon and remains economically meaningful after trading costs.

L4. Hosted-model and visual reliability.. Temperature zero and fixed prompts do not guarantee deterministic hosted-model outputs, and provider-side revisions or inference nondeterminism may still affect repeated runs. The paired design controls this variation within each Full/No-Alpha comparison, but not necessarily across independent executions.

The Pattern and Trend agents use deterministic OHLCV rules to detect obvious inconsistencies in visual conclusions. These rules check output consistency only and leave the VLM's actual use of the cited candle, wick, breakout, or trendline unverified. Prior work reports strong trend sensitivity and persistent spatial-reasoning limitations in financial chart understanding [11, 10]. Establishing causal visual grounding would require matched counterfactual charts and human-labelled local evidence, neither of which is included in this study.

L5. Sentiment coverage and attribution.. Cutoff filtering is an intended software control, not independent verification of archived inputs. Historical caches lack scorer provenance and are treated as unavailable in the verified-score numerical comparator, rather than as neutral sentiment. The three-symbol legacy ablation has unresolved branch provenance and does not establish independent sentiment effects or interactions. New provenance fields cannot retroactively identify the scorer or authenticate the missing historical requests.

7.2. Future Work

F1. Larger and longer evaluation.. The most important extension is a preregistered study over a broader universe such as the VN100 and multiple market regimes. More test points would increase per-symbol power, support sector-level analysis, and permit out-of-time validation rather than repeated analysis of the same recent period.

F2. Stronger baselines and multiplicity control.. Future experiments should add random-walk, logistic, tree-based, and sequence-model baselines on the identical information set. A time-ordered nested design should repeat normalisation, IC orientation, and top-five selection wholly inside each inner training fold, then score the frozen selected bundle on an untouched outer interval [17]. White's Reality Check, false-discovery-rate control, or the Deflated Sharpe Ratio should additionally account for searching across 85 candidates. Selection frequency, IC-sign agreement, and rank correlation across rolling calibration windows should be published rather than reporting only the winning factors.

F3. Execution and portfolio realism.. A portfolio simulator should model lot sizes, taxes, dynamic spreads, liquidity limits, partial fills, and market impact, and should allocate capital jointly across symbols. Confidence gates and risk-based sizing can then be assessed without conflating forecasting accuracy with all-in binary allocation.

F4. Component reliability.. Repeated hosted-model runs should quantify decision variance and archive provider revision metadata whenever available. ViSoBERT experiments should pin an exact Hub commit and test agreement between the local Transformers and Dedicated Endpoint paths before pooling their outputs. Vision faithfulness should be tested with two matched interventions: preserve global trend while changing only the local candlestick evidence, and preserve the local pattern while swapping the trend confounder. Directional changes should then be compared with rule-implied changes and human local-evidence annotations, rather than scored only against the future return. Sentiment should likewise be evaluated against a Vietnamese financial-domain test set. The observed negative VCB interaction also motivates an explicit conflict-resolution mechanism between factor and sentiment evidence.

F5. Per-factor translation audit.. Each registered adaptation should be profiled by horizon-specific IC decay, autocorrelation at several lags, sign persistence, exposure turnover, and net performance after the same two-leg costs used by the account simulator. Diagnostics should be stratified by symbol and market regime and compared with simple price/volume baselines. Factors whose orientation changes frequently or whose apparent IC disappears before $L = 3$ should be excluded by a preregistered stability rule rather than retained through ad hoc interpretation.

7.3. Broader Impact

Localized financial AI can broaden research beyond English-language, highly liquid markets, but automated use in a shallow market may amplify noise and herding. The framework should therefore be treated as decision-support research rather than trading advice. Reproducible releases must preserve model and data provenance while respecting news copyright, provider terms, and privacy obligations.

8. Conclusion

This reanalysis documents the evidence limits of a five-agent decision-support implementation for Vietnamese equities. The legacy Full branch achieves 65.0% raw accuracy, below the 68.9% constant DOWN/CASH benchmark, and has a negative mean account return. Balanced accuracy of 55.0% and MCC of 0.113 describe modest class-balanced discrimination in this single archived run, without establishing superiority or a reliable alpha effect. The weaker Base reports control also removes sentiment, so its comparison with Full cannot identify alpha information value.

The numerical reconstruction adds train-fitted majority, momentum, two logistic feature sets and factor-only baselines, with explicit preprocessing and daily ledgers. However, 74 of the 180 cycles differ from legacy and two individual labels change. These results therefore remain a separate exploratory analysis. Archived Full training predictions and complete report/feature inputs are still unavailable for a genuine Full-exposure-matched and identical-information comparison. Predictive superiority, isolated alpha/sentiment value and profitability are not established. Recovering the authenticated historical inputs or performing a newly frozen common-snapshot evaluation is required before those claims can be reconsidered; software checks and revised presentation alone do not complete that evidentiary requirement.

CRediT authorship contribution statement

Van-Tang Nguyen: Conceptualization (C); Software (So); Validation (Va); Investigation (I); Original draft preparation (O); Project administration (P); Funding acquisition (Fu).

Hoang-Dung Phan: Methodology (M); Software (So); Formal analysis (Fo); Investigation (I); Resources (R); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Visualization (Vi); Supervision (Su).

Thanh-Tam Do Thi: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Supervision (Su).

Phuong-Thai Nguyen: Methodology (M); Investigation (I); Data curation (D); Original draft preparation (O); Visualization (Vi); Supervision (Su).

Hong-Viet Tran: Conceptualization (C); Methodology (M); Software (So); Validation (Va); Formal analysis (Fo); Data curation (D); Original draft preparation (O); Writing-review and editing (E); Project administration (P).

CONFLICT OF INTEREST STATEMENT (mandatory)

Authors state no conflict of interest.

CODE AVAILABILITY

The source code and demonstration implementation of the proposed framework are available in the project repository.

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Appendix A. Agent System Prompts

This appendix presents the system and user prompts deployed in the agents.

Note: The original prompts in the codebase are written entirely in Vietnamese to match the target market's language conventions and to elicit native-level analysis from the LLMs. For the purpose of this paper, they have been translated into English and condensed, while strictly preserving their original logic, constraints, and instructions.

Appendix A.1. Decision Agent Prompt

The Decision Agent receives all upstream reports condensed to save tokens. The prompt enforces a forced binary direction forecast (LONG or SHORT), explicitly distinct from an executed action, and a structured reasoning process. The forecast horizon is dynamically injected as a bar count (e.g., `L=3 bars` for daily data), never as settlement notation. Below is the prompt template (with placeholder variables denoted by `{...}`):

```
You are a Senior Quant Trader with 20 years of experience in the VN stock market.
Asset: **{stock_name}** | Timeframe: **{time_frame}**.

MARKET RULE: {h_note}

TIME-INDEX CONTRACT: the input window is [e-W,e); the final visible decision
bar is d=e-1; candidate entry is at the next-bar open O[e]; the immutable
entry is O[e] and the target exit is C[e-1+L]. The label is UP only
when this LONG cycle remains profitable after costs on both legs.

CORE OBJECTIVE: Deeply evaluate reports from independent AI Agents
to forecast **{h_desc}**. You MUST make a definitive decision:
choose either **LONG** or **SHORT** (NEUTRAL is STRICTLY FORBIDDEN).

---
### [1] TREND ANALYSIS (Trend Agent)
{trend_report}
---
### [2] PRICE PATTERNS (Pattern Agent)
{pattern_report}
---
### [3] TECHNICAL INDICATORS (Indicator Agent)
{indicator_report}
---
### [4] ALPHA FACTORS & ORDER FLOW (Alpha Agent)
{alpha_report}
### [5] NEWS & MARKET SENTIMENT (Sentiment Analysis)
{sentiment_report}
---
## REASONING GUIDE (Chain of Thought)
1. Market Context (Trend)
2. Price Action & Timing (Pattern)
3. Momentum (Indicator)
4. Flow & Sentiment (Alpha + Sentiment)

## MANDATORY OUTPUT FORMAT
‘‘‘json
{
  "decision": "<LONG or SHORT>",
  "forecast_horizon": "{h_val}",
  "confidence": "<Very High | High | Medium | Low>",
  "risk_reward_ratio": <float>,
  "evidence_for": "<3 sentences summarizing reasons>",
  "evidence_against": "<Biggest risk>",
  "justification": "<Brief justification>"
}
‘‘‘
```

The agent reports 4 (when Alpha Agent is active) or 3 (No-Alpha ablation) upstream summaries. Sections `[4]` and `[5]` are omitted entirely when `include_alpha=False`.

Appendix A.2. Pattern Agent Prompt

The frozen configuration names the vision-language model `qwen/qwen3.8-27b`; the following Pattern Agent prompt is prototype source, not a provenance-validated research artifact:

```
/no_think
You are an AI specializing in Price Action pattern analysis.

ABSOLUTE RULES:
1. ONLY return the 6 fields below. Do NOT add any other fields.
2. NO reasoning, NO explanations.
3. NO fallback scenarios. Provide exactly ONE definitive assessment.
4. Each field on ONE line. Be concise.

MANDATORY FORMAT:
**Pattern:** <Name>
**Confidence:** High | Medium | Low
**Forecast Bias:** Bullish | Bearish | Neutral
**Evidence:** <Brief description>
**Key Candle:** <List key candle>
**Trade Implication:** <Scenario>
```

The 16 reference patterns provided to the model include: Inverse Head and Shoulders, Double Bottom, Falling/Rising Wedge, Triangle, Bull/Bear Flag, Rectangle, V-Reversal, and Clear Trend.

If the vision model fails, the agent falls back to a text-only analysis of the most recent 20 OHLCV rows using the text LLM (`openai/gpt-oss-20b`).

Appendix A.3. Trend Agent Prompt

The Trend Agent also utilizes a vision-language model to evaluate dynamically generated support and resistance lines.

```
You are an expert at analyzing K-line trend patterns in high-frequency trading.
Your sole objective: PREDICT {h_desc}.
You must always reply in the exact markdown format requested.
NEVER write all fields on a single line.

This is a {time_frame} candlestick chart with automatic trendlines:
- **Blue line** = support line (derived from close prices)
- **Red line** = resistance line (derived from close prices)

Your tasks:
1. Analyze how price interacts with the blue and red lines.
```

```
2. Determine if candles are bouncing, breaking out, or compressing.
3. Evaluate the slope and distance between the two lines.
4. Predict the direction for **{h_val}**: bullish, bearish, or sideways.

MANDATORY FORMAT:
**Trend Direction:** Bullish | Bearish | Sideways
**Support Level:** <specific price>
**Resistance Level:** <specific price>
**Trendline Slope:** Rising | Falling | Flat
**Price vs Support:** <bouncing / breaking / compressing>
**Detailed Analysis:** <2-3 sentences of price action analysis>
**Trend Prediction:** <1-2 sentences forecasting {h_desc}>
**Confidence:** High | Medium | Low
```

Appendix A.4. Indicator Agent Prompt

The Indicator Agent receives a pre-calculated table of technical indicators and their deterministic classifications (computed in Python). Its prompt forces it to adhere strictly to the system's hardcoded signals rather than re-interpreting the raw data.

```
You are an expert at High-Frequency Trading technical analysis.
SOLE OBJECTIVE: Predict price direction for **{h_desc}**.

MOST IMPORTANT RULES:
1. SIGNAL CLASSIFICATION and CONSENSUS have already been determined
   by the system - you MUST use these exact conclusions. Do NOT count
   or change them.
2. Each indicator has a strict [END ... SECTION] boundary. Do NOT use
   numbers from one section to describe another.
3. Your only task: write a clear, actionable qualitative interpretation
   for {h_short}.
4. NO hallucinated numbers. NO saying 'missing information'.

## Data Table (Calculated & Classified)
[Pre-rendered Table and System Consensus injected here]

## Requirement: Write an interpretation report
For EACH indicator, use this exact template:
**[Indicator Name]**
- {h_short} Signal: [EXACT system classification]
- Key levels/crossovers: [Specific numbers from the correct section]
- Momentum: [accelerating / decelerating / stable] - short explanation

Then summarize the final consensus:
**Signal Convergence Summary - Prediction for {h_short}**
- Bullish agreement: **{n_tang}/{total}**
```

```
- Bearish agreement: **{n_giam}/{total}**
- Primary Trend: **{consensus}** - Confidence: **{confidence}**
- Conclusion: [1-2 sentences summarizing potential movement]
```

Appendix A.5. Sentiment LLM Prompt (Subroutine of Alpha Agent)

Sentiment Analysis is invoked programmatically as a subroutine within the Alpha Agent node rather than acting as a standalone agent in the DAG. ViSoBERT runs through local Transformers by default and can instead use a configured Hugging Face Dedicated Inference Endpoint; a deterministic lexicon is the explicit last-resort fallback. The actual scorer, backend, and fallback flag are retained per newly scored article; the text LLM only synthesizes a qualitative summary of the resulting news bias and related-company impacts.

```
You are an expert at analyzing sentiment in the VN stock market.
Synthesize data from press articles. Reply in markdown.

Analyze sentiment for **{target}** ({time_frame}).

=== AGGREGATE RESULTS ===
Article Count: {article_count} | Avg Score: {avg_score} | Label: {label}
Distribution: {positive} / {neutral_count} / {negative}
Sentiment Bar: {sentiment_bar_visual} | Model: {model_note}

=== LATEST 15 ARTICLES ===
{art_sum}

=== RELATED COMPANIES ===
{rel_sum}

---
Write the report using this template:

**Sentiment Summary - {target}**
**Main Takeaway:** [1-2 sentences]
**Detailed Analysis:**
- News Trend: [...]
- Positive Signals: [...]
- Negative Signals: [...]
- Pressure from Related Companies: [...]
```

Appendix A.6. Alpha Agent LLM Reasoning Prompt

After computing the five alpha factors programmatically, the Alpha Agent invokes the text LLM to synthesise a qualitative interpretation. The prompt template is:

```
You are a quantitative and order flow analyst forecasting {horizon_label}.
Below are OHLCV-only alpha factors and a separate sentiment context
for **{stock_name}**.

### ALPHA FACTORS RESULTS
{report_md}

### SEPARATE SENTIMENT & NEWS CONTEXT
{sentiment_context_md}

FUSION CONTRACT:
- Alpha values and ranks use OHLCV only.
- Sentiment does not enter factor formulas; it is text context only.

Compare the two evidence sources qualitatively to:
1. Provide a short comment (1-2 sentences) on current momentum.
2. Synthesize: Are flow and sentiment in agreement or contradiction?
3. Identify hidden risks/opportunities from news not captured by technical alphas.

Do NOT over-analyze. Deduce from the numbers to conclude the view for {horizon_label}.
```

Appendix B. LangGraph State Schema

The entire multi-agent pipeline shares a single typed state object defined as a Python `TypedDict`. Table B.14 lists all fields, grouped by the agent that writes them and annotated with their data types. The backtest wrapper records the zero-based provenance separately for each test point: exclusive boundary e, window $[e-W, e)$, decision $d = e-1$, entry e, target $e-1+L$, and the source timestamp for every index. These fields make calendar gaps (including weekends and exchange holidays) explicit rather than inferring timestamps by adding calendar days.

Graph topology.. The DAG is constructed in `graph_setup.py` using the LangGraph `StateGraph` API. In the Full System configuration, the sequential edge order is:

$$\texttt{START} \rightarrow \texttt{Indicator} \rightarrow \texttt{Alpha} \rightarrow \texttt{Pattern} \rightarrow \texttt{Trend} \rightarrow \texttt{Decision} \rightarrow \texttt{END}$$

During evaluation, graph execution is split into a variant-independent upstream graph and variant-specific decision graphs. Indicator, Pattern, and Trend run once per exact cutoff; the resulting state is deep-copied into Full, Alpha-only, Sentiment-only, and Baseline branches as required by the experiment. This shared-artifact topology prevents Alpha settings from changing the No-Alpha evidence snapshot.

Appendix C. Proprietary Alpha Formula Implementation

This appendix provides the exact Python implementation of the five proprietary alpha formulas listed in Table 2. All formulas operate on the technical variables extracted from the OHLCV window by the function `_extract_tech_vars()`.

Table B.14: Complete LangGraph state schema (`IndicatorAgentState`)

Field	Type	Description
Input fields (set before graph execution)		
`kline_data`	`dict`	OHLCV dictionary for technical indicator computation
`interval`	`str`	Canonical market-data interval code (e.g., "1d")
`forecast_horizon_bars`	`int`	Target horizon L; label at decision $d = e - 1$ is $\log(C_{e-1+L}/C_{e-1})$
`holding_policy`	`str`	Explicit next-bar-open-to-target-close execution policy
`alpha_history`	`dict`	Point-in-time OHLCV prefix ending at decision cutoff; no entry/target bars
`alpha_cutoff`	`str`	Exact timestamp of the final observable decision bar
`history_requirements`	`dict`	Separate chart and alpha-calibration requests; calendar lookback and actual-bar minimums are distinct
`window_end_date`	`str`	Decision cutoff timestamp used for point-in-time sentiment availability
`time_frame`	`str`	Time period for K-line data (e.g., "1 ngày")
`stock_name`	`str`	Stock ticker symbol
`is_backtest`	`bool`	Flag to block external crawlers in backtest mode
`pattern_image`	`str`	Base64-encoded candlestick chart PNG
`trend_image`	`str`	Base64-encoded trendline-annotated chart PNG
Indicator Agent outputs		
`rsi`	`List[float]`	RSI values (period 14)
`macd, macd_signal, macd_hist`	`List[float]`	MACD line, signal, histogram (12, 26, 9)
`stoch_k, stoch_d`	`List[float]`	Stochastic %K, %D (14, 3, 3)
`roc`	`List[float]`	Rate of Change (period 10)
`willr`	`List[float]`	Williams %R (period 14)
`bb_upper, bb_middle, bb_lower`	`List[float]`	Bollinger Bands (20, 2)
`adx, plus_di, minus_di`	`List[float]`	ADX and directional indicators
`indicator_report`	`str`	Markdown summary with consensus
Alpha Agent outputs		
`alpha_report`	`str`	Five alpha factors + LLM reasoning
Pattern Agent outputs		
`pattern_report`	`str`	Six-field markdown pattern analysis
Trend Agent outputs		
`trend_report`	`str`	Eight-field markdown trend analysis
Sentiment outputs (written by Alpha Agent internally)		
`sentiment_report`	`str`	CafeF sentiment summary
`sentiment_data`	`Dict[str, Any]`	Display articles plus complete compact cutoff-safe observations
`sentiment_norm`	`Dict[str, Any]`	Time-decayed sentiment, relative context, availability and provenance
Decision Agent outputs		
`final_trade_decision`	`str`	Raw LLM response containing JSON decision
`decision_prompt`	`str`	Full prompt sent to the Decision Agent
`messages`	`List[BaseMessage]`	LangChain message history

Appendix C.1. α_1: Flow-Driven Momentum (FDM)

```
def _alpha1_fdm(tv: dict) -> dict:
    roc1_adj  = tv["roc1_adjusted"]    # ROC(1) / rolling_std(ROC, 20)
    vol_surge = tv["vol_surge"]        # Volume / ADV(20)
    raw   = roc1_adj * vol_surge * 0.5
    value = math.tanh(raw)
    # Signal threshold: |value| > 0.10
```

Where `roc1_adjusted` is the one-period rate of change normalised by its 20-period rolling standard deviation, and `vol_surge` is the ratio of current volume (or intra-bar range proxy when volume is unavailable) to the 20-period average volume.

Appendix C.2. α_2: MACD-Flow Proxy (MFP)

```
def _alpha2_mfp(tv: dict) -> dict:
    roc1_adj = tv["roc1_adjusted"]
    macd_hist = tv["macd_hist_cur"]
    z_macd = tanh(macd_hist / (close * 0.005)) * 1.5
    raw = z_macd + roc1_adj * 0.5
    value = math.tanh(raw)
```

This factor always uses price-derived MACD histogram and ROC inputs. Sentiment coverage does not switch or modify its formula; sentiment is provided only to the later LLM text-context comparison.

Appendix C.3. α_3: Liquidity Void Reversion (LVR)

```
def _alpha3_lvr(tv: dict) -> dict:
    z_close_dev5 = tv["zscore_close_dev5"]
    raw   = -1.0 * z_close_dev5 * 0.8
    value = math.tanh(raw)
    # Signal threshold: |value| > 0.15 (higher than other alphas)
```

Where `zscore_close_dev5` is the z-score of the series $\frac{C_u - \text{SMA}_5}{C_u}$ over a 20-period rolling window at bar position u. The negative sign creates a mean-reversion signal: when the close is abnormally far above SMA(5), the alpha predicts a pullback.

Appendix C.4. α_4: Bollinger Squeeze & Flow (BFE)

```
def _alpha4_bfe(tv: dict) -> dict:
    bb_pct_b = tv["bb_pct_b"]         # Bollinger %B
    vol_z5   = tv["vol_zscore_5"]     # Volume z-score (5-period)
    raw   = (bb_pct_b - 0.5) * vol_z5 * 1.5
    value = math.tanh(raw)
```

This formula fires when price breaks above/below the Bollinger midline ($\%B \neq 0.5$) with confirming volume expansion ($|Z_{\text{vol}}| > 0$). The multiplicative interaction ensures both conditions must co-occur.

Appendix C.5. α_5: Order Flow Exhaustion (OFE)

```
def _alpha5_ofe(tv: dict) -> dict:
    roc1_adj  = tv["roc1_adjusted"]
    close_pos = tv["close_pos"]   # (Close - Low)/(High - Low) - 0.5
    raw   = roc1_adj * close_pos * 3.0
    value = math.tanh(raw)
```

Where `close_pos` $\in [-0.5, 0.5]$ measures where the closing price falls within the intra-bar range, centred at zero. A positive `close_pos` with positive momentum confirms buying pressure; the $\times 3.0$ amplification increases signal responsiveness.

Appendix D. Indicator Signal Classification Logic

Section 4 describes the Indicator Agent's deterministic signal classification. This appendix provides the exact classification logic implemented in Python (function `_classify_signals`), which is injected into the LLM prompt as immutable ground truth.

Appendix D.1. MACD Classification

1. **Priority 1 — Crossover detection**: If $\text{MACD}_{t-1} < \text{Signal}_{t-1}$ and $\text{MACD}_t \geq \text{Signal}_t$, classify as BULLISH (Golden Cross). Conversely, classify as BEARISH (Death Cross).
2. **Priority 2 — Histogram direction**: If no crossover occurred, classify based on histogram sign and trend (expanding or contracting).

Appendix D.2. RSI Classification

- RSI ≤ 30: BULLISH (oversold bounce expected)
- RSI ≥ 70: BEARISH (overbought pullback expected)
- $55 \leq$ RSI < 70: BULLISH (positive momentum)
- $30 <$ RSI ≤ 45: BEARISH (negative momentum)
- Otherwise: NEUTRAL

Appendix D.3. Stochastic Oscillator Classification

1. **Priority 1 — %K/%D crossover**: %K crossing above %D → BULLISH; %K crossing below %D → BEARISH.
2. **Priority 2 — Extreme zones**: $\%K \geq 80 \rightarrow$ BEARISH (overbought); $\%K \leq 20 \rightarrow$ BULLISH (oversold).
3. **Otherwise**: Compare current %K vs %D position.

Appendix D.4. Williams %R Classification

- $\%R \geq -20$: BEARISH (overbought, near 0)
- $\%R \leq -80$: BULLISH (oversold, near -100)
- Neutral zone: Classify based on 3-bar trend direction (improving toward $0 \rightarrow$ BULLISH; deteriorating toward $-100 \rightarrow$ BEARISH).

Appendix D.5. ROC Classification

- ROC $> +0.5\%$: BULLISH
- ROC $< -0.5\%$: BEARISH
- Otherwise: NEUTRAL

Appendix D.6. Consensus Rule

The overall consensus is determined by majority vote over the five indicators:

- ≥ 4 agreement → High confidence
- 3 agreement → Medium confidence
- Otherwise → Low confidence

Appendix E. Software Dependencies and Reproducibility

Table E.15 lists the core software dependencies used in the implementation. All packages are specified in `requirements.txt` and are installable via `pip`.

Table E.15: Software dependencies

Package	Version	Purpose
`flask`	latest	Web interface for real-time demo
`pandas`	latest	Data manipulation and OHLCV processing
`numpy`	latest	Numerical computation
`matplotlib`	latest	Chart rendering backend
`mplfinance`	latest	Candlestick chart generation
`scipy`	latest	Statistical functions
`TA-Lib`	latest	Technical indicator computation (MACD, RSI, etc.)
`langchain`	latest	LLM abstraction layer
`langchain-groq`	latest	Groq API integration for LLM inference
`langgraph`	latest	Multi-agent DAG orchestration framework
`vnstock`	$\geq 3.0.0$	Vietnamese stock market data (VCI/MSN sources)
`requests`	latest	HTTP client for CafeF crawling and optional Dedicated Endpoint inference
`transformers`	latest	Local ViSoBERT model and fast-tokenizer pipeline
`torch`	latest	Local CPU/CUDA tensor inference
`sentencepiece`	latest	ViSoBERT tokenizer compatibility support
`beautifulsoup4`	latest	HTML parsing for article content extraction
`Pillow`	latest	Image processing utilities

Sentiment scorer provenance.. The runtime no longer sends this model to the deprecated generic Hugging Face serverless route. With the default automatic backend policy, it calls the exact Dedicated Endpoint URL supplied through the environment only when one is configured; otherwise it loads `5CD-AI/Vietnamese-Sentiment-visobert` locally. A failed dedicated endpoint in automatic mode also falls back to this local path. Local inference loads the Transformers sequence-classification model and the repository's fast `tokenizer.json`, rather than the incompatible legacy slow XLM-R tokenizer declaration, and applies truncation at 256 tokens. The official class mapping is `LABEL_0/NEG` (negative), `LABEL_1/POS` (positive), and `LABEL_2/NEU` (neutral). Only a ViSoBERT failure causes the deterministic lexicon fallback. New records store `scorer`, `scorer_backend`, and `is_fallback`.

The supplied historical cache schema stores label, numeric score, and confidence but predates those provenance fields. The audit therefore does not infer a scorer from numerical values: every such record is classified as `legacy-unknown`, and it contributes to a verified ViSoBERT or lexicon count only when explicit metadata is present. This conservative rule prevents a runtime configuration from being treated as evidence about an older article.

Web result serialization.. Background analysis can return NumPy or Pandas scalars even after all six agent steps have completed. Before Flask status endpoints call `jsonify`, the result payload is now normalized recursively to native JSON-compatible Python types. NumPy Boolean, integer, and floating scalars are converted to their Python counterparts; arrays and tabular vectors become lists; timestamps use ISO 8601; and non-finite floating values become JSON `null`. This transport-layer normalization prevents a completed job from returning HTTP 500 solely because a nested `numpy.bool_` or related object is not serializable. It does not alter the decision, sentiment score, or

archived experimental metric.

Table E.16: Ticker-selection strata and point-in-time sentiment coverage.

Ticker	Sector	Design stratum	Dated/total	V/L/U	Mean 90d	≥ 8 points
BHN	Food/beverages	State-associated, specialised	20/20	20/0/0	1.15	0/20
CMG	Information technology	Mid-cap	20/20	20/0/0	0.85	0/20
FPT	Information technology	VN30/large-cap	20/20	20/0/0	1.45	2/20
HVN	Aviation and tourism	State-associated	20/20	20/0/0	9.45	19/20
MBB	Banking	VN30/large-cap	20/20	20/0/0	7.35	7/20
MWG	Retail	VN30/large-cap	20/20	20/0/0	6.05	8/20
VCB	Banking	VN30/large-cap	20/20	20/0/0	9.90	18/20
VJC	Aviation and tourism	Large private carrier	20/20	20/0/0	10.10	17/20
VNM	Consumer food	VN30/large-cap	20/20	20/0/0	5.60	2/20
All	**Nine-symbol purposive sample**		**180/180**	**180/0/0**	**5.77**	**73/180**

Notes: V/L/U denotes verified ViSoBERT, verified lexicon, and unknown scorer counts. "Mean 90d" is the mean number of dated articles in the inclusive 90-day lookback at the 20 test cutoffs. The final column counts test points with at least eight eligible articles. The underlying machine-readable audit is generated by `scripts/analyze_sentiment_coverage.py`.

Reproducibility artifacts.. The following artifacts are provided or specified for reproducibility:

- **Model versions**: The reported text default is `openai/gpt-oss-20b`; the vision default is `qwen/qwen3.8-27b`. Both use the Groq API. The backtest engine resolves these configurations directly and stores the effective run settings with its result artifacts.
- **Temperature**: The current default is $T = 0.0$. Effective temperature, token caps, reasoning format, retry count, and a stable credential-free configuration hash are recorded per run; this metadata does not guarantee deterministic hosted-model output.
- **Retry policy**: The application policy defaults to at most three text calls (two retries with 5/10-second exponential waits) and two vision calls (one retry with a 3-second wait) when SDK retries are disabled. Positive SDK `max_retries` budgets make the SDK the single retry owner and reduce the application wrapper to one call; SDK-internal attempts are not observable and remain unknown in telemetry. Inter-test pacing is recorded separately from retry delays.
- **Historical sentiment cache**: Nine JSON files (`sentiment_cache_{SYMBOL}.json`), one per evaluated stock, contain pre-crawled and scored CafeF articles. Point-in-time filtering uses the parsed publication date; undated records are rejected in strict research mode. The supplied files lack scorer metadata, so all 180 cached scores are labelled `legacy-unknown` in the coverage audit.
- **Default configuration**: Stored in `default_config.py` with all hyperparameters.

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